

dirac16complex and the dark sector: pressure, energy density and equation of state from numerical solutions

Five CVODE experiments, their independent verification, and whether the framework is connected to dark matter or dark energy

dirac16complex project (with Claude Opus 5.5)

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Abstract

This is the Stage-3 document of the `dirac16complex` project. It asks how the pressure and the energy density of `dirac16complex` behave and change in gravitational backgrounds, and whether the framework has any connection to dark matter or dark energy. `dirac16complex` is the complex, Grassmann-odd, 16-component spinor of $\text{Spin}(4,4)$ of Stages 1 and 2. Five experiments were solved as ordinary differential equations with the pure-Rust SUNDIALS 7.8.0 CVODE engine of the `rustSolveIt` repositories. EXP-1 puts the field in the primordial pair-creation background of the author's notebook. EXP-2 is a self-consistent homogeneous 8-dimensional Einstein cosmology sourced by a mean-field condensate. EXP-3 is a 4-dimensional effective late universe in which the condensate plays the role of dark energy, compared with the Unite supernova fits quoted in the reference PDF, $(w_0, w_a) = (-0.861, -0.60)$ and $w = -0.764$. EXP-4 follows a thermal Fermi gas of `dirac16complex` quanta and their gravitational pair creation by the transition from inflation to radiation. EXP-5 follows the modes with momentum along the extra times. All 69 self-checks of the Rust program pass, as do all 162 checks of five independent numpy checkers (including repeat-run byte identity and refined-tolerance convergence), all 10 checks of the EXP-3 analysis, all 71 assertions of a Jupyter notebook and all 49 checks of a Mathematica notebook that re-derives every reduced equation symbolically and re-integrates the solutions with `NDSolve`. **Dark matter:** in the sector without extra-time momentum the quanta are ordinary massive fermions. Their gas has $w = 0.3329$ at temperature $T = 10m$ and $w = 0.0359$ after the scale factor has grown a hundredfold (on the momentum grid $k \leq 12T$; 0.0361 without the cut), and expansion creates them for $m > 0$ (comoving number na^3 between 1.45×10^{-3} and 4.99×10^{-3} in units of H_{inf}^3 , mostly in the radiation era when H is of order m) but not for $m = 0$. That is a dark-matter-like equation of state and a production mechanism, not a dark-matter model: the abundance, the darkness, and the stabilisation of the extra dimensions that dust-like dilution in 3-space needs were neither computed nor derived. **Dark energy:** negative pressure arises only from an attractive four-fermion interaction. Tuned to the Unite value $w_0 = -0.861$, the condensate has $w_a = -4.81$, its effective mass vanishes and $w = -1$ at redshift $z = 0.026$, and its adiabatic sound speed squared today is -12.4 . Beyond that redshift the single mean-field mode occupies a negative-energy level, so the phantom epoch, the negative energy density beyond $z = 0.293$ and the bounce ($H = 0$) at $z = 0.388$ that the equations then give are artefacts of the mean field, not established properties; no state of positive-energy quanta of this Lagrangian is phantom. Deflating extra times give p/ρ the tangent CPL parameters of the Unite fit, but the CPL fit of that p/ρ over the data range is $(-0.640, -1.98)$, and they violate 4-dimensional energy conservation, imply $\dot{G}/G = 2.63 H_0$ and need a negative 8-dimensional energy density. Within everything computed here the framework does not provide viable dark energy, and it reproduces neither the Unite (w_0, w_a) nor $w = -0.764$ consistently. All results are homogeneous and mean-field; no perturbations and no data likelihood were computed.

1 The question and the short answer

The question of this stage is: *in detail, investigate, determine and discuss, using numerical solutions and graphs, the physical behaviour and changing nature of pressure and energy density in this framework; is there any connection to dark matter and/or dark energy?*

The short answer, which Section 12 argues in full:

- **What makes ρ and p change?** The 7-volume V (the condensate density obeys $SV = \text{const}$), the redshift of momenta, and the self-interaction λ .
- **Does anything behave like dark matter?** Yes in its equation of state: a Pauli-consistent gas of quanta goes from $w = 1/3$ to $w \rightarrow 0$, pairs are created gravitationally for $m > 0$, and

the single-mode mean-field condensate is dust (up to a Fermi-degeneracy pressure that was not computed).

- **Is it a dark-matter model?** No: the abundance, the darkness, the stabilisation of the extra dimensions and the perturbations are not established.
- **Does anything give $w < -1/3$?** Only an attractive four-fermion term ($\lambda < 0$), and, as a dilution effect only, the frozen density of the primordial field.
- **Does it fit Unite?** No: $w_0 = -0.861$ forces $w_a = -4.81$, and the mean field leaves its valid regime (positive-energy occupied level) at $z = 0.026$; beyond it the equations give a phantom epoch, negative energy density beyond $z = 0.293$ and a bounce at $z = 0.388$, which are artefacts of that approximation.
- **Deflating extra times?** Their tangent of p/ρ equals the Unite fit values, but the fitted CPL parameters of that p/ρ are $(-0.640, -1.98)$, and they break 4D energy conservation, the constancy of G_N and the 8D energy bound.

2 Scope and non-claims

Scope. The object is the Stage-1 field with the Lagrangian of Section 4.1. The backgrounds are homogeneous. Every computed number in this document is copied from a committed output: the summary.json files of the Rust study, the python-check-report.json files of the independent checkers, fits.json of the EXP-3 analysis, numerics-summary.json, notebook-report.json and mathematica-report.json (Section 14). The Unite values and the scalar-field formulas are quoted from the reference PDF (Section 3.3). Where a statement is derived here from those numbers or from verified closed forms rather than machine-checked, the text says so.

Non-claims.

1. No claim is made that `dirac16complex` is dark matter or dark energy. The computations show which equations of state the framework can produce and what goes wrong.
2. The signature (4,4) is not observed spacetime. EXP-3 and EXP-4 assume that the hidden space and the extra times are static (stabilised). That assumption is not derived; EXP-2 shows that 8-dimensional Einstein gravity with this matter does not stabilise them.
3. All condensate results are mean-field: the bilinears are evaluated on one c-number rest mode by the expectation-value rule of Section 4.6, at Hartree level, without the Pauli constraint on the occupation of that mode (Section 4.6), and they are valid only while that mode is a positive-energy level of the instantaneous Hamiltonian. The gas and pair-creation results are sums over free modes. No interacting quantum field theory, renormalisation or loop correction is computed.
4. No perturbations, no structure formation, no CMB, BAO or supernova likelihood and no parameter estimation were computed. The distance-modulus fits of EXP-3 are fits of noise-free model curves with equal weights.
5. The Unite numbers and the scalar-field formulas are taken from the reference PDF (Section 3.3), an e-mail that cites no primary publication; they were not checked against one and are not re-derived from data here.

3 Notation and conventions

3.1 Coordinates, units and matrices

- Everything is counted from 0: coordinates $x_0, \dots, x_7$, frame indices $a = 0, \dots, 7$, spinor components $\Psi_0, \dots, \Psi_{15}$. The tangent metric is $\eta = \text{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$: directions 0 to 3 are space-like, 4 to 7 time-like. x_4 is the evolution time t ; x_0 is the hidden space, x_1, x_2, x_3 are 3-space and x_5, x_6, x_7 are the three extra times.
- γ^a are the real integer 16 by 16 gamma matrices of the split-octonion (notebook) basis, $\{\gamma^a, \gamma^b\} = 2\eta^{ab}$. $C = \gamma^0\gamma^1\gamma^2\gamma^3$ is the charge (adjoint) matrix, $\bar{\Psi} = \Psi^\dagger C$, and $B = -iC\gamma^4$ is Hermitian with $B^2 = 1$ and eigenvalues ± 1 (8 each). $u^\dagger u$ is the Hilbert norm and $u^\dagger B u$ the Krein norm of a mode amplitude $u \in \mathbb{C}^{16}$.
- Units: EXP-1 uses $H = m = \kappa = 1$ with $t = Hx_4$; EXP-2 uses $m = \kappa_8 = 1$; EXP-3 uses $H_0 = c = 1$ and densities in units of $3H_0^2/\kappa_4$; EXP-4 uses $m = 1$ for the gas and $H_{\text{inf}} = 1$ for pair creation; EXP-5 uses $H = m = 1$.

3.2 Symbols that are used twice

- x_0 is the hidden coordinate, and in EXP-2 and EXP-3 also the dimensionless coupling $x_0 = \lambda S_0/(2m)$ (the reports call it x_0). The coupling always appears as a number, for example $x_0 = -0.462654$.
- $z = 6Hx_0$ in the primordial field (Stage 2); z is the redshift in EXP-3.
- μ is the reduced spinor frequency of EXP-3; the distance modulus is written $\text{DM}(z)$.
- K is the hidden-space momentum in EXP-1 and the physical momentum k/a in EXP-4.
- q is the extra-time momentum of EXP-5; the deceleration parameter is written q_{dec} .
- γ is also the extra-time deflation index of the EXP-3 variant ($c \propto a^{-\gamma}$); it never carries an index there.
- H is the notebook's inverse length in EXP-1, H_i are Hubble rates, H_0 is today's Hubble rate and H_{inf} the de Sitter rate of EXP-4. $\varkappa(t)$ is the WKB growth rate of EXP-5.

3.3 The reference PDF, the CPL form and a sign discrepancy

The reference on the equation of state is the PDF in the repository root whose name begins with "Gmail - w = equation of state parameter" (an input, git-ignored and not committed). It treats a canonical scalar field with $\mathcal{L}_\phi = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ in signature $(+, -, -, -)$, for which $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V$, $P_\phi = \frac{1}{2}\dot{\phi}^2 - V$, $w_\phi = P_\phi/\rho_\phi$ and $\dot{\phi} + 3H\phi + V'(\phi) = 0$. Our conventions (mostly plus for space, $\mathcal{L} = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V$) give the same ρ and p . The PDF states that a canonical field has $w \geq -1$ and that crossing the phantom divide $w = -1$ needs non-canonical physics.

The PDF uses the Chevallier-Polarski-Linder (CPL) form $w(a) = w_0 + w_a(1 - a)$. It quotes the Unite supernova compilation (Pantheon+ and DES-SN5YR combined): the time-evolving fit $w_0 = -0.861$ (also written ≈ -0.86), $w_a = -0.60$, hence $w_0 + w_a = -1.461$ in the deep past (phantom); and a

constant- w (wCDM) fit $w = -0.764$ of the supernovae alone, "roughly two standard deviations" from -1 . No error bars on (w_0, w_a) are given in the PDF. It lists the Λ CDM components as radiation $w = 1/3$, matter $w = 0$, dark matter $w = 0$ and the cosmological constant $w = -1$. The PDF calls -0.764 a statistical hint, not a definitive value. Its citation markers have no reference list, so the Unite values used here are those quoted in the e-mail; they were not checked against a primary publication. The PDF gives no value of Ω_m ; the value $\Omega_m = 0.305$ of EXP-3 is an input of the numerical programme.

Sign discrepancy in the PDF. From its own formula, $dw/da = -w_a$. Since a grows with time, a field whose w starts near -1 and increases (thawing) has $w_a < 0$, and a field whose w approaches -1 from above (freezing) has $w_a > 0$. The PDF's table of thawing and freezing models writes the opposite: thawing " $(w_a > 0)$ " and freezing " $(w_a < 0)$ ". This document uses the formula-consistent signs throughout: **thawing means $w_a < 0$, freezing means $w_a > 0$** . The Unite fit ($w_a = -0.60$) and every attractive dirac16complex condensate of EXP-3 have the thawing sign of w_a (w increases with time), but neither is a thawing field in the sense of the PDF and of the standard classification: a thawing field starts frozen at $w \approx -1$ and stays at or above -1 , whereas both start phantom ($w < -1$) and cross $w = -1$.

4 The framework

4.1 Field and Lagrangian

$\Psi = (\Psi_0, \dots, \Psi_{15})^T$ has complex Grassmann-odd components and is a section of the complex spinor bundle of $\text{Spin}(4,4)$. It couples to gravity through the vielbein e_μ^a and the canonical spin connection $\Omega_\mu = \frac{1}{2}\omega_{\mu ab}S^{ab}$, $S^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$, with $D_\mu\Psi = \partial_\mu\Psi + \Omega_\mu\Psi$ (Stage 1). The Lagrangian is

$$\mathcal{L} = \sqrt{|g|} \left[\frac{1}{2}(\bar{\Psi}\gamma^\mu D_\mu\Psi - (D_\mu\bar{\Psi})\gamma^\mu\Psi) - m\bar{\Psi}\Psi - U(\bar{\Psi}\Psi) \right],$$

$$\bar{\Psi} = \Psi^\dagger C, \quad S = \bar{\Psi}\Psi, \quad U(S) = \frac{\lambda}{2}S^2, \quad M_{\text{eff}} = m + U'(S) = m + \lambda S.$$

There is no factor i ($C\gamma^a$ is real antisymmetric) and no cosmological constant: for a Grassmann field U must be a polynomial with $U(0) = U'(0) = 0$. $\lambda < 0$ is an attractive and $\lambda > 0$ a repulsive four-fermion contact interaction.

4.2 Field equations

$$\gamma^\mu D_\mu\Psi = (m + \lambda S)\Psi, \quad (D_\mu\bar{\Psi})\gamma^\mu = -(m + \lambda S)\bar{\Psi}.$$

The spin connection cannot be gauged away in a curved field: $(\gamma^\mu D_\mu)^2\Psi = g^{\mu\nu}(D_\mu D_\nu\Psi - \Gamma^\lambda_{\mu\nu}D_\lambda\Psi) - \frac{1}{4}R\Psi$ (Stage 1, Lichnerowicz coefficient $-1/4$). For a diagonal vielbein $e_\mu^a = \text{diag}(h_0, \dots, h_7)$, $\gamma^\mu\Omega_\mu = \frac{1}{2}\sum_b h_b^{-1}\partial_b \ln(\prod_{c \neq b} h_c) \gamma^b$.

4.3 Energy-momentum tensor, energy density and pressures

$$T_{\mu\nu} = -\frac{1}{4} \left[\bar{\Psi} \gamma_\mu D_\nu \Psi + \bar{\Psi} \gamma_\nu D_\mu \Psi - (D_\mu \bar{\Psi}) \gamma_\nu \Psi - (D_\nu \bar{\Psi}) \gamma_\mu \Psi \right] + g_{\mu\nu} \mathcal{L}_s,$$

$$\mathcal{L}_s = \mathcal{L} / \sqrt{|g|} \stackrel{\text{on shell}}{=} SU'(S) - U(S), \quad T^\mu{}_\mu \stackrel{\text{on shell}}{=} -mS + 3\lambda S^2.$$

With Gaussian normal time ($g_{44} = -1$) and the observer $u = \partial_4$: $\rho = T_{44}$, $p_{(i)} = T^i{}_i$ (no sum) for the seven transverse directions $i \neq 4$ (four space-like, three time-like), $\bar{p} = \frac{1}{7} \sum_{i \neq 4} p_{(i)}$ and $w = p/\rho$ (or $\bar{w} = \bar{p}/\rho$ when the pressures differ).

4.4 Kinetic and potential energy, and the equation of state

Two exact splits are used, as in Stage 1. With $K_4 = \frac{1}{2}(\bar{\Psi} \gamma^{x_4} D_4 \Psi - (D_4 \bar{\Psi}) \gamma^{x_4} \Psi)$ and $K_\perp$ the other seven terms of the kinetic term:

- **(A) Lagrangian split**, the exact analogue of the scalar field's $\frac{1}{2}\dot{\phi}^2$ and V : $\text{KE}_L = \frac{1}{2}K_4$, $\text{PE}_L = \rho - \text{KE}_L$.
- **(B) Hamiltonian split**: $\text{KE}_H = -K_\perp$ (momentum or gradient energy), $\text{PE}_H = mS + U$ (rest mass and interaction), and $\rho = \text{KE}_H + \text{PE}_H$ identically.

For a homogeneous isotropic condensate on shell:

$$\rho = mS + \frac{\lambda}{2}S^2, \quad p = \frac{\lambda}{2}S^2, \quad w = \frac{\lambda S}{2m + \lambda S} = \frac{\text{KE}_L - \text{PE}_L}{\text{KE}_L + \text{PE}_L},$$

$$\text{KE}_L = \frac{1}{2}S(m + \lambda S), \quad \text{PE}_L = \frac{1}{2}mS, \quad \text{KE}_H = 0, \quad \text{PE}_H = \rho, \quad \rho + p = 2\text{KE}_L = S M_{\text{eff}}.$$

So in the single-mode mean field a free condensate ($\lambda = 0$) is dust ($\text{KE}_L = \text{PE}_L$, $w = 0$; Section 4.6 gives the Pauli caveat), a repulsive one is stiffer than dust ($0 < w < 1$), and an attractive one has negative pressure. For $\rho > 0$ the state is phantom ($w < -1$) exactly when $\text{KE}_L < 0$, that is when $S M_{\text{eff}} < 0$. $\text{KE}_L = \varepsilon/2 > 0$ for every positive-energy quantum: on positive-energy eigenmodes $s = M_{\text{eff}}/E$, so a state of such quanta has $S M_{\text{eff}} = \sum n M_{\text{eff}}^2/E \geq 0$, hence $\rho + p \geq 0$ and $\text{KE}_L \geq 0$. $\text{KE}_L < 0$ needs an occupied negative-energy level, which is what the mean-field mode becomes when M_{eff} changes sign (Section 4.6): no positive-energy (normal-ordered) state of this Lagrangian is phantom. For a gas of free quanta, $\text{KE}_L = \text{PE}_L = \rho/2$ for every mode, so the relation $p = \text{KE} - \text{PE}$ of split (A) holds only for homogeneous condensates; the Hamiltonian split then gives $\text{KE}_H = 3p$ and $\text{PE}_H = \rho - 3p$.

4.5 Homogeneous backgrounds, dilution and the mode equation

All experiments use a homogeneous diagonal metric with lapse 1,

$$ds^2 = -dt^2 + \sum_{i \in T} \epsilon_i h_i(t)^2 dx_i^2, \quad T = \{0, 1, 2, 3, 5, 6, 7\}, \quad \epsilon_i = +1 \ (i \leq 3), \ -1 \ (i \geq 5),$$

with Hubble rates $H_i = \dot{h}_i/h_i$, expansion $\Theta = \sum_i H_i$ and 7-volume $V = \prod_i h_i$. The groups are $b = h_0$ (hidden space), $a = h_1 = h_2 = h_3$ (3-space) and $c = h_5 = h_6 = h_7$ (extra times), so $\Theta = H_b + 3H_a + 3H_c$ and $\gamma^\mu \Omega_\mu = \frac{1}{2}\Theta\gamma^4$. A mode $\Psi = V^{-1/2}e^{ik \cdot x}u(t)$ obeys

$$i\dot{u} = h(t)u, \quad h(t) = -iM_{\text{eff}}\gamma^4 - \gamma^4 \sum_{j \in T} \frac{k_j}{h_j} \gamma^j.$$

h is Hermitian exactly when $k_5 = k_6 = k_7 = 0$. Per mode, $\varepsilon(u) = u^\dagger h u$, $p_j(u) = -(k_j/h_j) u^\dagger \gamma^4 \gamma^j u$ (equal to $(k_j/h_j)^2/E$ on eigenmodes) and $s(u) = u^\dagger (-i\gamma^4)u$ (equal to M_{eff}/E on positive-energy eigenmodes). For the condensate ($k = 0$) h is a multiple of $-i\gamma^4$, so $s(u)$ is conserved and $S \propto 1/V$ exactly, for any $M_{\text{eff}}(t)$; Stage 1 derives the same law from the continuity equation $\partial_4 \rho + \sum_i H_i(\rho + p_{(i)}) = 0$ with $\rho + p = SM_{\text{eff}}$ wherever $M_{\text{eff}} \neq 0$. **This dilution law drives almost everything below:** $mS \propto V^{-1}$ behaves like dust, while $\frac{\lambda}{2}S^2 \propto V^{-2}$ dominates at small volume and dies away at large volume.

4.6 Quantization, the good sector and the expectation-value rule

Canonical quantization with respect to x_4 gives $\{\Psi, \Psi^\dagger\} = B\delta^7/\sqrt{|g|}$ in Gaussian normal gauge. The canonical state space is a Krein space of signature (8,8) per mode. In the sector without extra-time momentum (the good sector) the single-particle Hamiltonian is Hermitian, the dispersion is $E^2 = m^2 + k_0^2 + \dots + k_3^2$, and there is an ordinary Fock space with particles and antiparticles of energy $\sqrt{m^2 + k^2}$ above a filled Dirac sea. Modes with extra-time momentum have $E^2 = m^2 + \dots - k_5^2 - k_6^2 - k_7^2$, which can become negative (Section 10). The unitarily implemented symmetry of the positive structure is $\text{Spin}(4) \times \text{Spin}(3)$, generated by the 9 compact S^{ab} that commute with B ; the 4 boosts S^{a4} ($a \leq 3$) also commute with B but are Hermitian, not unitary (Stage 1, erratum E1).

Expectation-value rule. In the positive-norm quantization a one-particle state built on a normalised positive-energy mode u above the Dirac sea has $\langle \Psi^\dagger M \Psi \rangle = u^\dagger B M u$, normal ordered relative to the sea. Hence the scalar density is $s(u) = u^\dagger B C u = u^\dagger (-i\gamma^4)u$, equal to $+1$ on every positive-energy rest eigenvector. The c-number reading $u^\dagger C u$ is a different quantity: on the positive-energy rest eigenspace $C = B$ exactly, so $u^\dagger C u$ is the Krein sign ± 1 of a joint eigenvector. The two readings agree for $B = +1$ (the states of EXP-2 and EXP-3) and differ in sign for $B = -1$ (for example the EXP-1 runs pos_Bm). The mean-field condensate is $S = S_0(V_0/V) s(u)$.

Limits of the mean field. Two limits matter for the conclusions.

1. **The occupied level must stay a positive-energy level.** At $k = 0$, $h = -iM_{\text{eff}}\gamma^4$ only changes the phase of u , so $s(u) = 1$ stays fixed while M_{eff} may change sign; then $u^\dagger h u = M_{\text{eff}} s(u) < 0$ and the single occupied mode has become a negative-energy eigenvector of the instantaneous h . That is outside the domain of the rule, which is stated for a quantum above the sea. The EXP-2 and EXP-3 rows with $M_{\text{eff}} < 0$, which contain every phantom row, every row with $\rho < 0$ and the approach to the EXP-3 bounce, are therefore mean-field artefacts of fixed normal ordering, not established properties of the field. Normal ordering against a fixed sea is least reliable exactly at the gap closure $M_{\text{eff}} = 0$, where the sea modes stop evolving adiabatically. (The same hole reading is applied to the negative-energy runs of EXP-1 in Section 6.5.)

2. **The Pauli principle.** The condensate is one rest mode multiplied by a macroscopic density. For a Grassmann field at most 8 positive-energy quanta can occupy one momentum state, so a finite density of fermions cannot all sit in the $k = 0$ mode. A Pauli-consistent homogeneous state at density n is a degenerate Fermi sea whose Fermi momentum k_F grows with n and which has a degeneracy pressure. The dust behaviour of the free condensate is therefore a property of the single-mode picture: a real state is dust only up to a Fermi-degeneracy pressure, negligible when $k_F(n) \ll m$. This was not computed. By an order-of-magnitude estimate (not a machine-checked result) it is negligible for EXP-3 if the physical mass is of order eV or more; for the EXP-2 densities it depends on the unspecified sizes of the extra dimensions and is not guaranteed. The λ term is also treated at Hartree level only, $U(\langle S \rangle)$; for a contact interaction the Fock exchange term is of the same order and is not included. The Pauli-consistent result of this document is the gas of EXP-4 ($f \leq 1, 8 + 8$ states per momentum).

4.7 dirac16complex beside a canonical scalar field

Quantity	Scalar field (PDF)	dirac16complex
energy density	$\frac{1}{2}\dot{\phi}^2 + V$	$mS + \frac{\lambda}{2}S^2$
pressure	$\frac{1}{2}\dot{\phi}^2 - V$	$\frac{\lambda}{2}S^2$
kinetic energy	$\frac{1}{2}\dot{\phi}^2 \geq 0$	$\text{KE}_L = \frac{1}{2}SM_{\text{eff}}$, never negative for quanta of positive energy
potential energy	V	$\text{PE}_L = \frac{1}{2}mS$
$w \rightarrow -1$	$\dot{\phi}^2 \ll V$	$M_{\text{eff}} \rightarrow 0$
phantom $w < -1$	impossible for $\rho > 0$	only with an occupied negative-energy level (mean-field artefact)
dilution	depends on $V(\phi)$	$S \propto 1/V$ always
cosmological constant	$V = \text{const}$	absent ($U(0) = 0$)

5 Numerical method, software and verification strategy

5.1 Engine and pinned commits

Every ordinary differential equation below is integrated by CVODE of the pure-Rust SUNDIALS 7.8.0 engine (sundials_rs) from the rustSolveIt repositories of once-ere; nothing is stepped by hand. The setup scripts setup_solver.ps1 and setup_solver.sh in scripts/ clone the platform repository at a pinned commit into the git-ignored vendor/rustSolveIt:

```
https://github.com/once-ere/rustSolveIt_Win11_SUNDIALS_7_8_0
a8fdff459adfe181573d7924b18bffbdf378fdb3
https://github.com/once-ere/rustSolveIt_macos-silicon_SUNDIALS_7_8_0
5360157f4f6160978f66400566c31b2ae25dd44d
https://github.com/once-ere/rustSolveIt_linux_SUNDIALS_7_8_0
6f58e02e53717a51375bd4bc5918edc57088d922
```

All three vendor the same sundials_rs. The engine is BSD-3-Clause and is fetched at build time, not redistributed.

Two CVODE configurations are used (src/driver.rs of the study):

- **BDF + Newton + dense direct linear solver** with CVODE’s difference-quotient Jacobian (EXP-1).
- **Adams-Moulton + fixed-point (functional) iteration**, without a linear solver (EXP-2 to EXP-5). This is the classical choice for non-stiff oscillatory problems: the Jacobian $-ih$ of the mode equation has imaginary eigenvalues in the good sector. EXP-4 selects it by a recorded test against BDF (Section 9.4).

5.2 The Rust study

The crate studies/dirac16complex_cosmology (edition 2021, unsafe code forbidden, warnings denied, its own workspace, compiled with the target feature +fma set in .cargo/config.toml) has this command line:

```
dirac16complex_cosmology <print-config|exp1|exp2|exp3|exp4|exp5|all>
  [--output DIR] [--rtol X] [--atol X] [--refined]
```

The refined option divides rtol and atol by 10 and max_step by 2. Following the planet_Mercury contract of rustSolveIt, every self-check prints a PASS or FAIL line and the last stdout line is SUCCESS or FAILURE. The gamma matrices, C , γ^8 and B are generated from the Stage-1 algebra fixture into src/generated.rs, which records the fixture sha256. Outputs are deterministic: CSV with a header row, 17 significant digits, LF line ends, and one summary.json per experiment with parameters, tolerances, solver counters, the fixture hash, every check and the verdict; no timing or absolute path is written. The outputs live in the directories exp1 to exp5 under artifacts/dirac16complex/numerics.

Solver work over the five canonical runs: 2.897×10^8 steps and 4.374×10^8 right-hand-side evaluations, almost all of them in the 48 thermal modes and the 64 pair-creation modes per mass of EXP-4.

5.3 Independent checkers, determinism and convergence

For each experiment an independent checker (numpy and the standard library only) rebuilds the gamma matrices from the fixture and recomputes every physical quantity from the raw spinor columns of the CSV files, never from the derived columns. It checks finite-difference residuals of the right-hand side, exact or closed-form solutions and invariants, and two further properties:

- **repeat-run byte identity**: it reruns the binary into a fresh directory and requires every output file to be byte-identical;
- **refined-tolerance convergence**: it reruns with the refined option and requires the errors to shrink or, where a floor is identified, to stay within it.

An analysis script does the EXP-3 fits (Nelder-Mead in numpy; no scipy) and writes fits.json with its own 10 validation checks, and a collector writes numerics-summary.json:

```

scripts/check_dirac16complex_exp1.py ... scripts/check_dirac16complex_exp5.py
scripts/analyze_dirac16complex_exp3.py
studies/dirac16complex_cosmology/tools/build_numerics_summary.py

```

Totals: 69 of 69 Rust self-checks, 162 of 162 checker checks and 10 of 10 analysis checks true; verdict SUCCESS.

5.4 Notebooks and figures

Two notebooks in notebooks/ re-examine the results independently (Section 11). The Jupyter notebook `dirac16complex_dark_sector.ipynb` is adapted from `rustSolveIt planet_Mercury/notebook: the build_notebook.py` helpers `md()` and `code()`, `find_binary()` and `run()` with the last-line SUCCESS contract, a gauntlet() of assertions, the headless standard-library executor `run_notebook.py` and the auditor `check_notebook.py`. They were copied with attribution (BSD-3-Clause per the `rustSolveIt Cargo` manifests; author once-ere). The notebook produced the 17 figures in `artifacts/dirac16complex/numerics/figures`.

None of the three rustSolveIt repositories (Win11 a8fdff45, macOS 5360157f, Linux 6f58e02e) contains a Mathematica notebook (.nb, .wl or .wls): this was verified by listing their full git trees, and each has 294 Jupyter notebooks. The Mathematica notebook `Dirac16ComplexDarkSector.nb` is therefore new. It is modelled on `dirac-main's notebooks/DiracTriality.nb` and its build and verify scripts `build_mathematica_notebook.wls` and `verify_mathematica_notebook.wls` (RunProcess plus Import of CSV and RawJSON). It produced the 8 figures in `figures/mathematica`.

6 EXP-1: the frozen field in the primordial pair-creation background

6.1 Purpose

The primordial field of the notebook inflates 3-space while three extra times deflate and the 7-volume stays constant. EXP-1 asks what `dirac16complex` does there and what source 8-dimensional Einstein gravity would need to produce this field.

6.2 Equations solved

The background (Stage 2, $t = Hx_4$, $H = 1$) has scale factors e^{a_4} for 3-space and e^{-a_4} for the extra times (the constant factors $s^{1/6}$ dropped), so V is constant, with the primordial window

$$\begin{aligned}
a'_4(t) &= \frac{A}{4} \left(1 + \tanh \frac{t-t_1}{\Delta} \right) \left(1 - \tanh \frac{t-t_2}{\Delta} \right), & A \in \{1, 2\}, \quad t_1 = 2, \quad t_2 = 7, \quad \Delta = 0.5, \\
a_4(t) &= \frac{A\Delta [\ell(2(t-t_1)/\Delta) - \ell(2(t-t_2)/\Delta)]}{2(1 - e^{-2(t_2-t_1)/\Delta})}, & \ell(x) = \ln(1 + e^x).
\end{aligned}$$

In the sector without 3-space and extra-time momentum, a hidden-space plane wave in the proper hidden coordinate $\zeta = \ln(\sin z)/(6H)$ reduces the Dirac equation (verified in Stage 2 and again symbolically in the Mathematica notebook) to

$$\Psi = e^{-3H\zeta} e^{iK\zeta} u(t) : \quad \gamma^4 \dot{u} = (M_{\text{eff}} - iK\gamma^0) u \quad \Longleftrightarrow \quad i\dot{u} = hu, \quad h = -iM_{\text{eff}}\gamma^4 - K\gamma^4\gamma^0.$$

The reduction is exact for $\lambda = 0$. The mode's scalar density is $S = e^{-6H\zeta} S_0 s(u) = S_0 s(u)/\sin z$, which depends on x_0 ; with $\lambda \neq 0$ the effective mass $M_{\text{eff}} = m + \lambda S$ therefore varies with x_0 , and the reduced equation with a constant $M_{\text{eff}} = m + \lambda S_0 s(u)$ holds only pointwise, as a fixed- x_0 mean-field approximation in which S_0 is the local density factor $S \sin z$ (Stage 2 says the same). CVODE integrates the 32 real components of u (state layout: re $u_0, \dots, u_{15}$, then im $u_0, \dots, u_{15}$). With the density factor S_0 and $M_{\text{eff}} = m + \lambda S$, $S = S_0 s(u)$:

$$\begin{aligned} \rho &= S_0 u^\dagger h u - \frac{\lambda}{2} S^2, & p_j &= S_0 p_j(u) + \frac{\lambda}{2} S^2, & \bar{p} &= \frac{1}{7} \sum_{j \in T} p_j, & w &= \bar{p}/\rho, \\ \text{KE}_L &= \frac{1}{2} S_0 u^\dagger h u, & \text{PE}_L &= \rho - \text{KE}_L, & \text{KE}_H &= S_0 \sum_j p_j(u), & \text{PE}_H &= mS + \frac{\lambda}{2} S^2. \end{aligned}$$

For constant M_{eff} , h is constant with $h^2 = E^2$, $E = \sqrt{M_{\text{eff}}^2 + K^2}$, and the exact solution is $u(t) = (\cos Et - i \sin Et h/E) u_0$. The source that 8D Einstein gravity $G^\mu{}_\nu = \kappa T^\mu{}_\nu$ would need is

$$\begin{aligned} p_{\text{req}} &= -\frac{3H^2(7 + a_4'^2)}{\kappa}, & p_{\text{req},0} &= -\frac{3H^2(a_4'^2 - 5)}{\kappa}, \\ p_{\text{req},1..3} &= \frac{H^2(15 - 3a_4'^2 + a_4'')}{\kappa}, & p_{\text{req},5..7} &= \frac{H^2(15 - 3a_4'^2 - a_4'')}{\kappa}. \end{aligned}$$

6.3 Initial data and parameters

$H = m = \kappa = 1$; $t \in [0, 10]$ with 200 output intervals (201 samples); hidden momenta $K \in \{0, 0.5, 2\}$; $S_0 = 1$. Four initial spinors per (A, K) : joint eigenvectors of h and B with (energy, Krein) signs $(+, +)$, $(+, -)$ and $(-, +)$, and a generic normalised mixture of both energy signs. Only the positive-energy eigenstates are physical one-particle states; the $(-, +)$ and mixed runs are numerical controls (Section 6.5). One extra run per profile has $\lambda = 0.5$, $K = 0.5$ (the pointwise mean-field approximation of Section 6.2): its initial state is the positive-energy eigenvector of the self-consistent mass $M_* = 1 + \lambda S_0 M_*/\sqrt{M_*^2 + K^2}$, $M_* = 1.47348266406420$. In total 26 runs.

6.4 Solver settings and statistics

BDF + Newton + dense linear solver, rtol 10^{-12} , atol 10^{-14} , max_step 0.02: 33866 steps and 34950 right-hand-side evaluations over the 26 runs. At rtol 10^{-10} the norm and energy drifts were about 4×10^{-8} , above the a priori conservation limit 10^{-8} ; the tolerance was tightened rather than the limit relaxed.

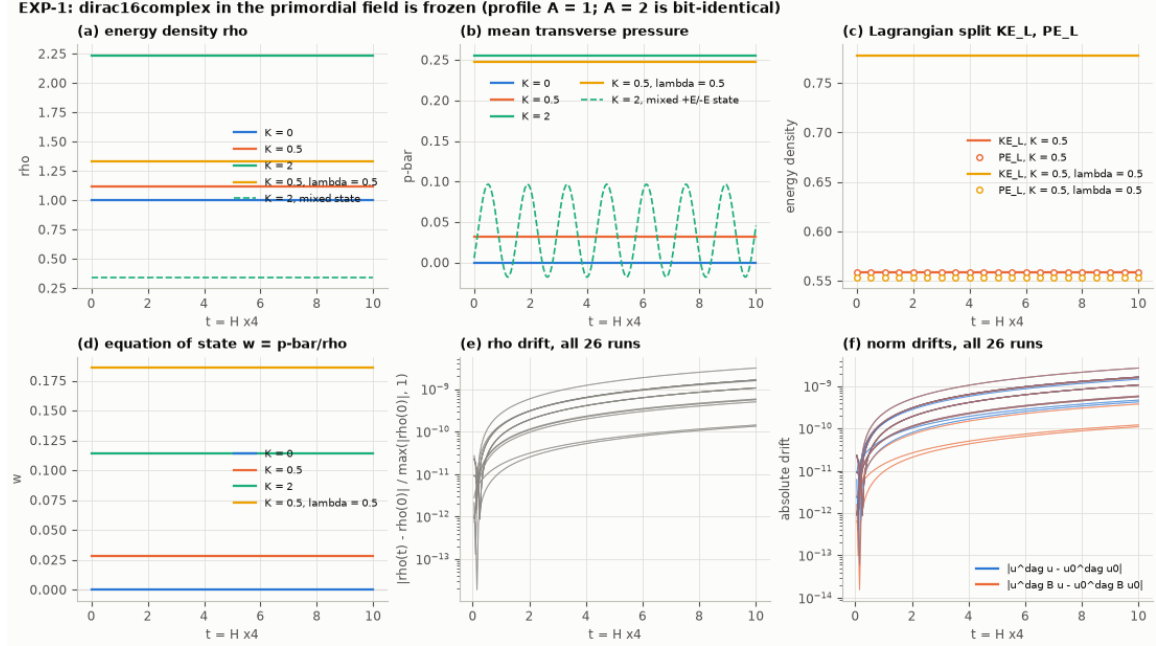


Figure 1: EXP-1 observables. (a) energy density, (b) mean transverse pressure, (c) Lagrangian split, (d) equation of state $w = \bar{p}/\rho$, (e) and (f) drifts of ρ and of the Hilbert and Krein norms over all 26 runs. Profile $A = 1$ is shown; $A = 2$ is bit-identical.

6.5 Results

- Frozen observables.** $\rho = S_0 u^\dagger h u$ is constant for every initial spinor (maximum drift 4.19×10^{-9}), because h is constant and Hermitian. The pressures and S are constant for energy eigenstates (drift 2.64×10^{-9}). For the mixed state with $K \neq 0$ the hidden-space pressure p_0 and S oscillate at frequency $2E$ (interference of the $+E$ and $-E$ sectors, a Zitterbewegung): the range of p_0 is 0.803 for $K = 2$ and 0.341 for $K = 0.5$, and the checker's exact solution reproduces the range to 3.7×10^{-10} . The mixed runs are numerical controls only, like the $(-, +)$ runs: their negative-energy components are levels that the sea fills in the positive-norm Fock space (Pauli-blocked), so $u^\dagger B M u$ is not the expectation value in any Fock state of such a spinor, and the $2E$ oscillation is a first-quantised interference effect, not physics of the frozen field. The mixtures take negative values of p_0 , $\bar{p}$ and S at some times (visible in their CSV columns), and they carry nonzero tensor bilinears $u^\dagger B C \gamma^i \gamma^j \gamma^4 u$, $i < j$, whereas the eigenstates do not; in the anisotropic primordial background those bilinears source off-diagonal stresses T_{ij} that the diagonal pressures leave out.
- Eigenmode laws.** For $K = 0$: $\rho = 1$, $p = 0$, $w = 0$, $\text{KE}_L = \text{PE}_L = 0.5$ (dust). For $K = 0.5$: $\rho = E = 1.1180$, $p_0 = K^2/E = 0.2236$, $w = K^2/(7E^2) = 0.02857$. For $K = 2$: $\rho = 2.2361$, $p_0 = 1.7889$, $w = 0.11429$. In the Lagrangian split every free eigenmode has $\text{KE}_L = \text{PE}_L$ ($w_L = 0$) whatever K is; the actual mean pressure is the hidden-space momentum flux.
- Repulsive interaction.** The $\lambda = 0.5$ run (pointwise, fixed- x_0 mean field) has $\rho = 1.3318$, $\bar{p} = 0.2471$, $w = 0.1856$, $\text{KE}_L = 0.7780$ and $\text{PE}_L = 0.5538$; the six pressures $p_1, \dots, p_7$ equal $\frac{\lambda}{2} S^2 = 0.2242$. M_{eff} stays at M_* to 1.3×10^{-9} .
- Profile independence.** a_4 enters h only through k_j/h_j with $k_j = 0$, so the $A = 1$ and $A = 2$

spinors are bit-identical (difference 0.0): no spurious a_4 coupling exists.

- **Negative-energy states.** The $(-, +)$ runs have $\rho = -E$ in this c-number reading; physically an antiparticle is a hole in such a mode and carries $+E$ (EXP-4). They are integrated as controls.

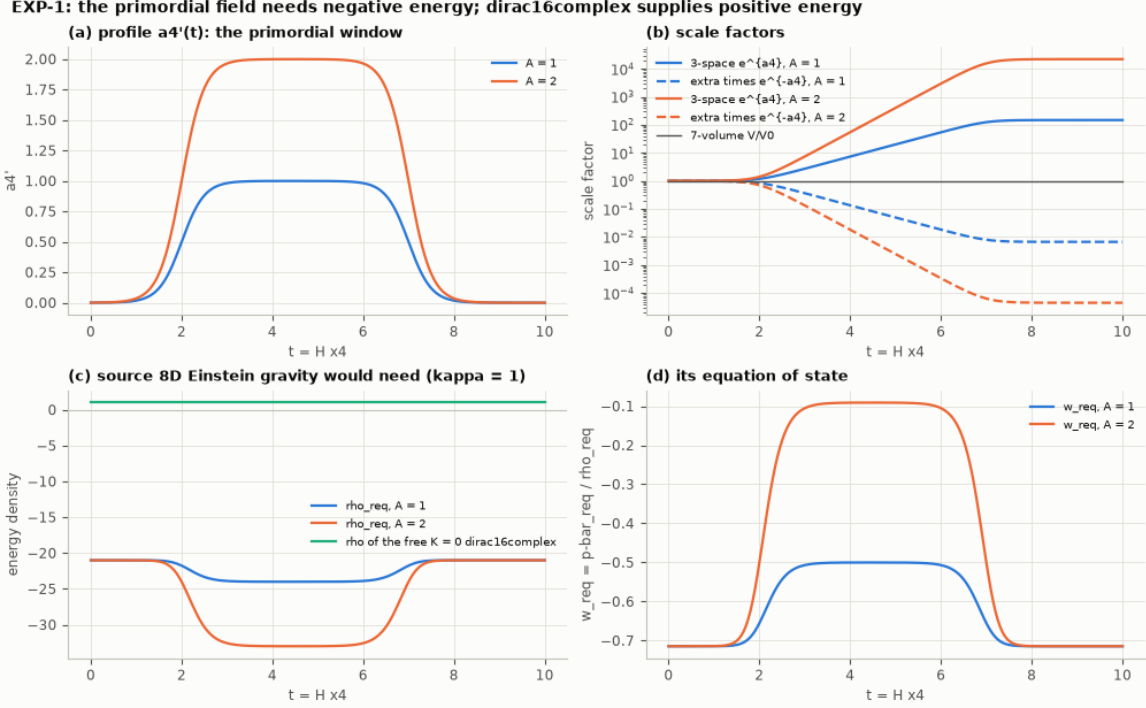


Figure 2: EXP-1 background and Einstein requirement. (a) the window $a_4'(t)$, (b) the 3-space factor e^{a_4} , the extra-time factor e^{-a_4} and the constant 7-volume, (c) ρ_{req} for both profiles beside the energy density of the free $K = 0$ field, (d) $w_{\text{req}} = \bar{p}_{\text{req}}/\rho_{\text{req}}$.

- **The required source has negative energy.** ρ_{req} lies in $[-24.0, -21.0]$ for $A = 1$ and in $[-33.0, -21.0]$ for $A = 2$; its maximum is $-21.000000000113 < 0$. The ratio w_{req} lies in $[-0.714, -0.500]$ for $A = 1$ and in $[-0.714, -0.091]$ for $A = 2$. It is negative because $\rho_{\text{req}} < 0$ while the mean required pressure is positive (all seven required pressures equal 15 where $a_4' = a_4'' = 0$), not because the pressure is negative. a_4 reaches 5.0 ($A = 1$) and 10.0 ($A = 2$), so 3-space grows by e^5 and e^{10} while $V/V_0 - 1$ stays at 6.7×10^{-16} .

6.6 Verification

The Rust program passes 10 of 10 self-checks and the EXP-1 checker 25 of 25. The checker recomputes the background, integrating a_4' by its own quadrature (1.1×10^{-14}); recomputes every derived column from the raw spinor (8.9×10^{-16}); evaluates a five-point finite-difference residual of $i\dot{u} = hu$ and of the literal reduced equation (worst ratio to the truncation bound 0.63) and of the kinetic term from the time derivative (0.31); compares with the exact propagator (maximum error 6.25×10^{-9} , limit 10^{-7}); and checks the frozen quantities, the eigenmode laws (1.9×10^{-9}), the

norms (drift 2.70×10^{-9}), the self-consistent mass, the profile independence, repeat-run byte identity, and refined convergence (every run's exact error strictly improved; largest refined-minus-canonical difference 6.13×10^{-9}). The Mathematica notebook derives the reduction symbolically, reproduces the Stage-2 Einstein tensor, re-integrates all 26 runs with an order-8 explicit Runge-Kutta NDSolve (maximum state deviation from Rust 2.92×10^{-9} ; its own exact error 4.3×10^{-14} against Rust's 6.25×10^{-9}) and passes a 9-point finite-difference residual test (1.04×10^{-10} relative over 5018 resolved rows) whose negative control with a flipped mass sign fails as it should (1.19).

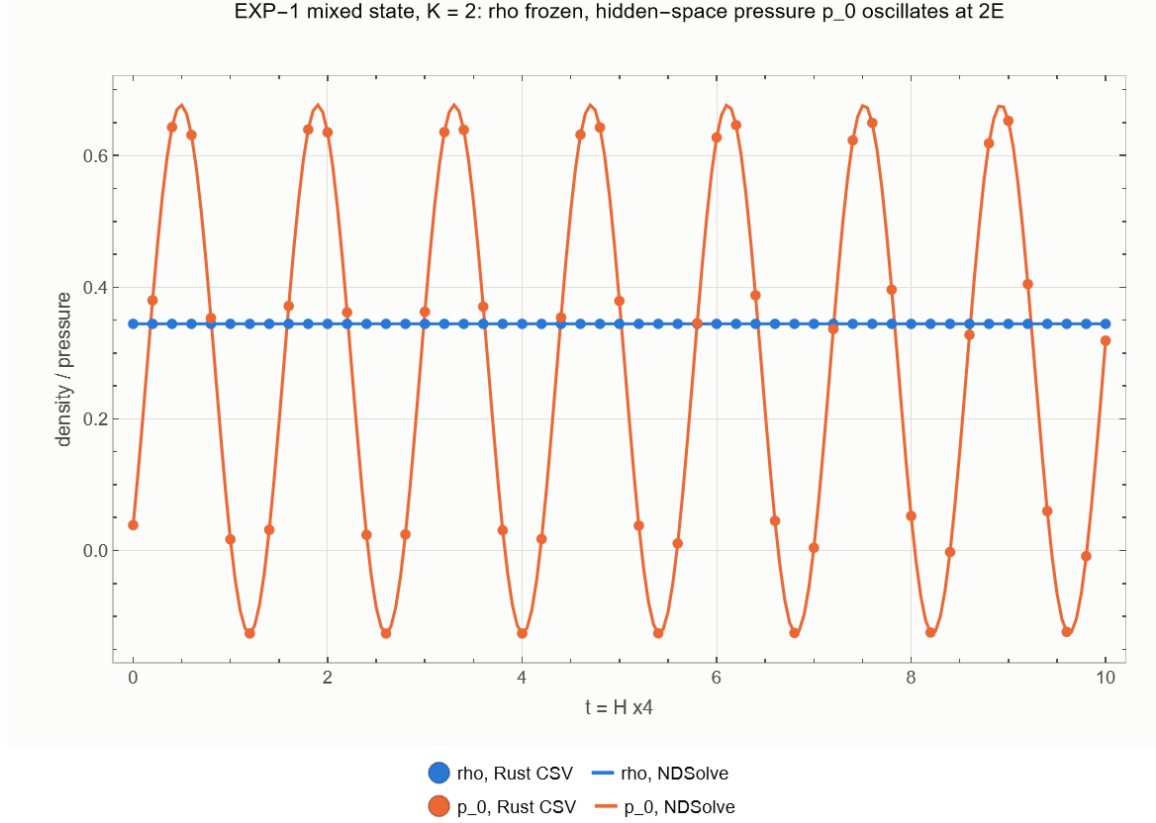


Figure 3: Mathematica cross-check of the mixed state with $K = 2$: ρ frozen and p_0 oscillating at $2E$; dots are the Rust CSV samples, lines the independent NDSolve solution.

6.7 Physical interpretation

In the primordial field the 7-volume is constant, so the dilution law $S \propto 1/V$ freezes the field: energy density and pressure do not change at all while 3-space inflates by up to e^{10} . (The $K = 0$ state is not a homogeneous condensate: its density goes as $1/\sin z$ along the hidden direction.) The oscillating pressures of the mixed runs are first-quantised interference of numerical controls (Section 6.5), not a property of the field. The frozen field's own equation of state is dust ($K = 0$) or slightly stiff ($K \neq 0$, or $\lambda > 0$ in the pointwise mean field); none of the positive-energy eigenstates has negative pressure. The geometry itself would need a source of negative energy density in 8D Einstein gravity, which positive-energy dirac16complex states cannot supply (Stage 2 found an exact source only for linear a_4 , and it too has negative energy). The frozen density matters for dark energy only as a dilution effect seen from 3-space (Section 12.3).

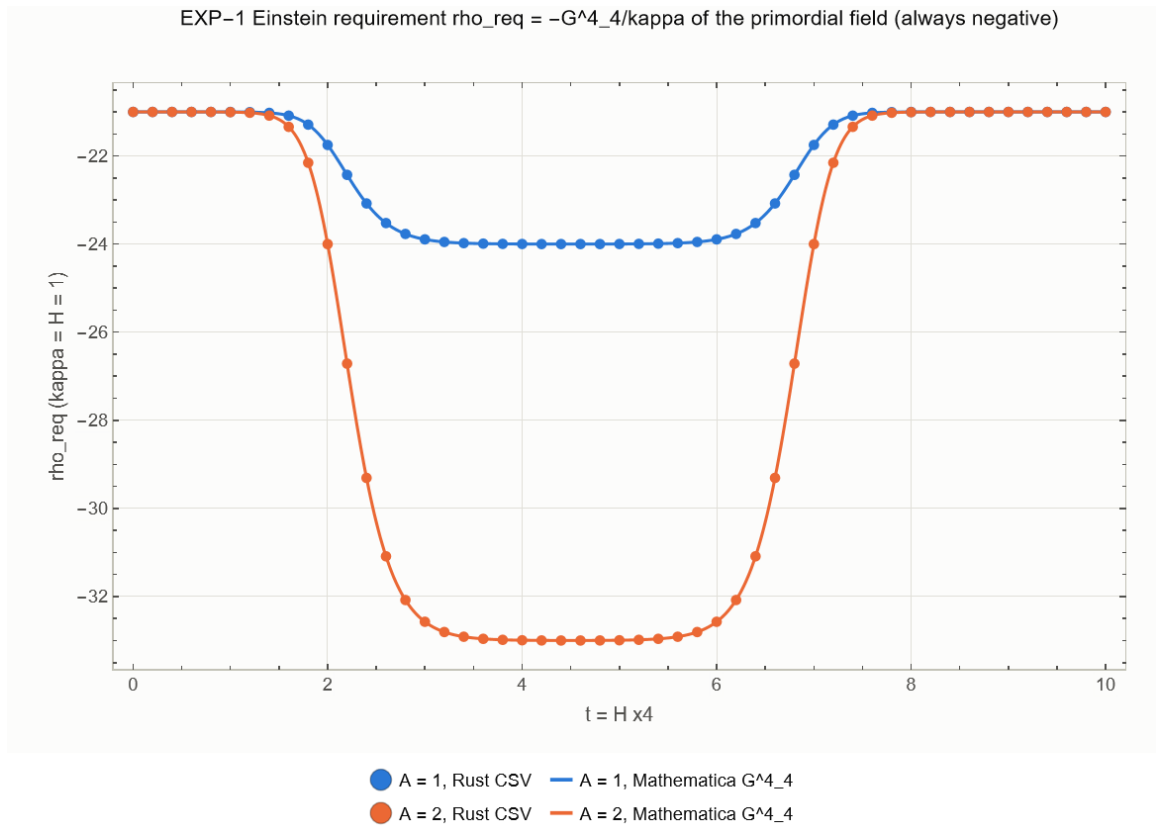


Figure 4: Mathematica cross-check of the Einstein requirement $\rho_{\text{req}} = -G^4_4/\kappa$ computed from the metric: always negative.

7 EXP-2: self-consistent 8-dimensional Einstein cosmology

7.1 Purpose

EXP-2 couples the condensate to 8-dimensional Einstein gravity with the hidden space, 3-space and extra times free, and asks how ρ , p and the anisotropy evolve, whether the extra times keep deflating, and what a 3-space observer would infer.

7.2 Equations solved

$\kappa = \kappa_8 = 1$, $ds^2 = -dt^2 + b^2 dx_0^2 + a^2(dx_1^2 + dx_2^2 + dx_3^2) - c^2(dx_5^2 + dx_6^2 + dx_7^2)$, $V = ba^3c^3$. The mixed Einstein tensor gives

$$\begin{aligned} \text{constraint: } & \sum_{i < j} H_i H_j = 3H_b H_a + 3H_b H_c + 3H_a^2 + 3H_c^2 + 9H_a H_c = \kappa\rho, \\ \text{evolution: } & \dot{H}_i = -H_i \Theta + \frac{\kappa}{6}(\rho - p), \quad \dot{h}_i = H_i h_i \quad (\frac{1}{6} = \frac{1}{D-2}, D=8), \\ \text{matter: } & i\dot{u} = hu, \quad h = -iM_{\text{eff}}\gamma^4, \quad S = S_0 \frac{V_0}{V} s(u), \quad \rho = mS + \frac{\lambda}{2}S^2, \quad p = \frac{\lambda}{2}S^2. \end{aligned}$$

CVODE integrates 38 reals: $\ln b$, $\ln a$, $\ln c$, H_b , H_a , H_c and the 32 components of u . Also recorded: $w = p/\rho$, $\text{KE}_L = \frac{1}{2}SM_{\text{eff}}$, $\text{PE}_L = \frac{1}{2}mS$, the bound $\Theta^2 - 3H_a^2 - 2\kappa\rho$ (equal to $H_b^2 + 3H_c^2 \geq 0$ on the constraint surface) and the dilution-inferred 3-space equation of state

$$w_{\text{eff}} = -1 + \frac{\Theta}{3H_a}(1 + w),$$

which is what an observer who assumes $\rho \propto a^{-3(1+w_{\text{eff}})}$ in 3-space would infer. It is a dilution measure, not the deceleration of 3-space.

Exact solution (derived in the study and re-derived by the checker). Because $\rho - p = mS$ does not depend on λ and SV is constant, $\ddot{V} = \frac{7}{6}\kappa mS_0$ and $(H_i V)^\cdot = \beta = \kappa mS_0/6$: $V(t) = 1 + \Theta_0 t + \alpha t^2$ with $\alpha = 7\beta/2$, $H_i = (H_{i0} + \beta t)/V$ and $u(t) = e^{-i \int M_{\text{eff}} dt} u_0$. Backward in time $V \rightarrow 0$ linearly at the Kasner-type singularity $t_s = -2/(\Theta_0 + \sqrt{D})$, $D = \Theta_0^2 - 4\alpha$, with exponents $p_i = (H_{i0} + \beta t_s)/\sqrt{D}$, $\sum p_i = 1$ and $\sum p_i^2 = 1 - 2\kappa m x_0 S_0/D$. Late times: $H_i t \rightarrow 2/7$ (isotropic 8D dust) and $w_{\text{eff}} \rightarrow 4/3$. The Rust code uses this closed form only for self-checks and to place the output times; CVODE integrates the full system.

7.3 Initial data and parameters

$m = 1$; $H_b = 0$, $H_a = 1$, $H_c = -0.2$ (extra times initially deflating), $\ln b = \ln a = \ln c = 0$; $u(0)$ is the first vector of the joint ($h = +M_{\text{eff}}$, $B = +1$) eigenspace. The constraint left-hand side is $C_0 = 1.32$ and fixes $S_0 = C_0/(1 + x_0)$, with $\lambda = 2mx_0/S_0$, $\rho(0) = 1.32$ and $M_{\text{eff}}(0) = m(1 + 2x_0) = 1, 0.2$ and 2 for the three runs:

Run	S_0, λ	α, β	t_s
$x_0 = 0$ (dust)	1.32, 0	0.77, 0.22	-0.4954
$x_0 = -0.4$ (attractive)	2.2, -0.3636	1.2833, 0.3667	-0.6266
$x_0 = +0.5$ (repulsive)	0.88, 1.1364	0.5133, 0.1467	-0.4624

The output grid is uniform in $\ln V$ with step 0.05 on $[-7, 18.5]$ (140 backward and 370 forward intervals, 511 rows). Backward the run stops at $V/V_0 = e^{-7}$, about 10^{-3} in t above t_s ; forward it ends at $V/V_0 = e^{18.5}$, $t_{\text{end}} = 11855.5, 9183.5$ and 14519.6 .

7.4 Solver settings and statistics

Adams + fixed point, rtol 10^{-12} , atol 10^{-15} , max_step 0.02: 1779784 steps and 1781396 right-hand-side evaluations. The step cap is needed: without it Adams takes steps of about 0.045 and accumulates an amplitude error of about 5×10^{-13} per step on the oscillating spinor, which drifts $s(u)$ by 1.7×10^{-7} over $t \sim 10^4$ and pushes the constraint residual to 3×10^{-8} .

7.5 Results

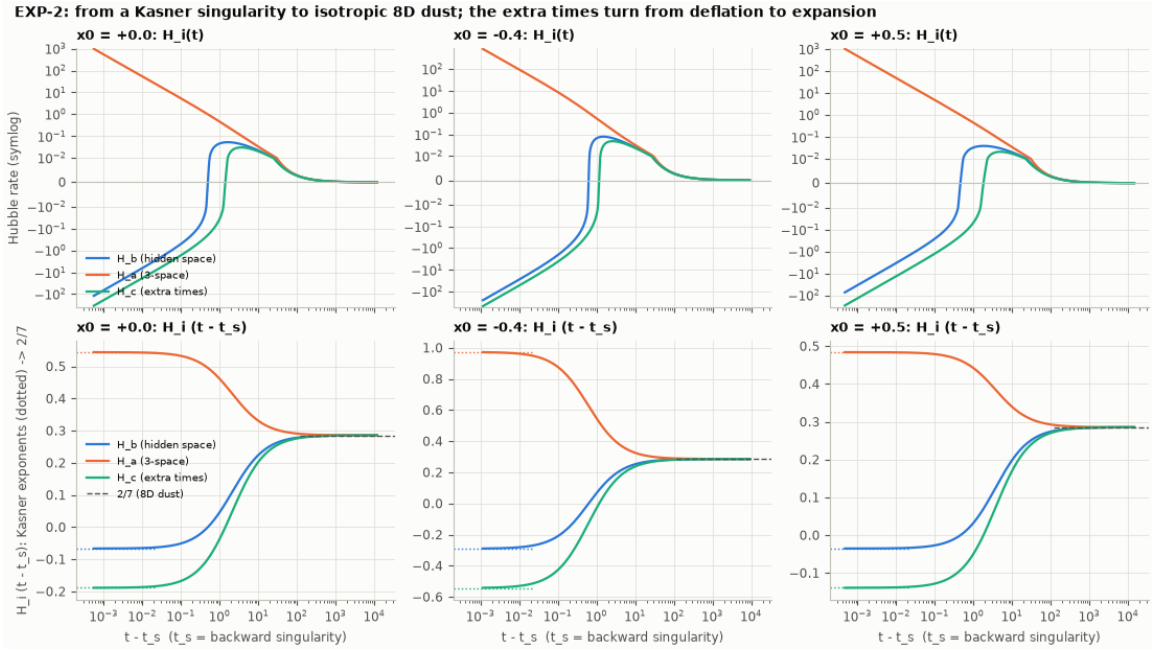


Figure 5: EXP-2 Hubble rates of the hidden space, 3-space and the extra times against $t - t_s$ (top, symmetric log scale), and $H_i(t - t_s)$ with the Kasner exponents dotted and the 8D dust value $2/7$ dashed (bottom), for the three runs.

- **Isotropisation.** The anisotropy $(H_i - H_j)V$ is conserved (drift 6.2×10^{-10}), so it decays as $1/V$. At the forward end the relative anisotropy $7 \max |H_i - H_j|/\Theta$ is 4.6×10^{-4} , 3.6×10^{-4} and 5.6×10^{-4} , $H_i t$ approaches $2/7 = 0.2857$, and $w_{\text{eff}} = 1.33275, 1.33288$ and 1.33261 approaches $4/3$.

- **The extra times turn from deflation to expansion** at $t = 0.2/\beta = 0.9091, 0.5455$ and 1.3636 (measured on the output grid: $0.9095, 0.5457, 1.3639$). Nothing in this dynamics keeps them deflating.
- **Kasner singularity.** The exponents (p_b, p_a, p_c) are $(-0.066576, 0.544271, -0.188746)$ with $\sum p_i^2 = 1$ (vacuum Kasner) for $x_0 = 0$; $(-0.290250, 0.972978, -0.542895)$ with $\sum p_i^2 = 3.80851$ for $x_0 = -0.4$; and $(-0.035225, 0.484182, -0.139107)$ with $\sum p_i^2 = 0.76259$ for $x_0 = +0.5$. The $\lambda S^2/2$ term is a stiff ($w = 1$) component near the singularity. Extrapolating the CVODE solution to $V = 0$ reproduces them to 2.9×10^{-7} .

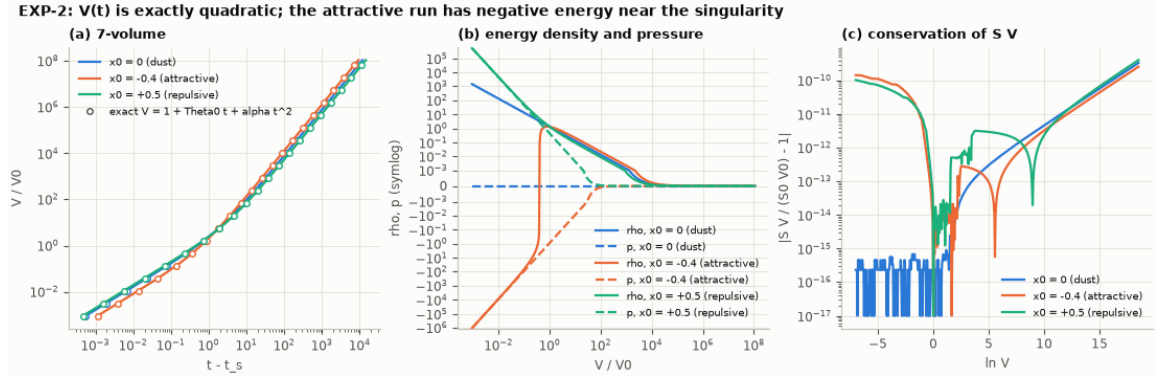


Figure 6: EXP-2 (a) the 7-volume against $t - t_s$ with the exact quadratic, (b) energy density and pressure against V/V_0 (symmetric log scale), (c) conservation of SV .

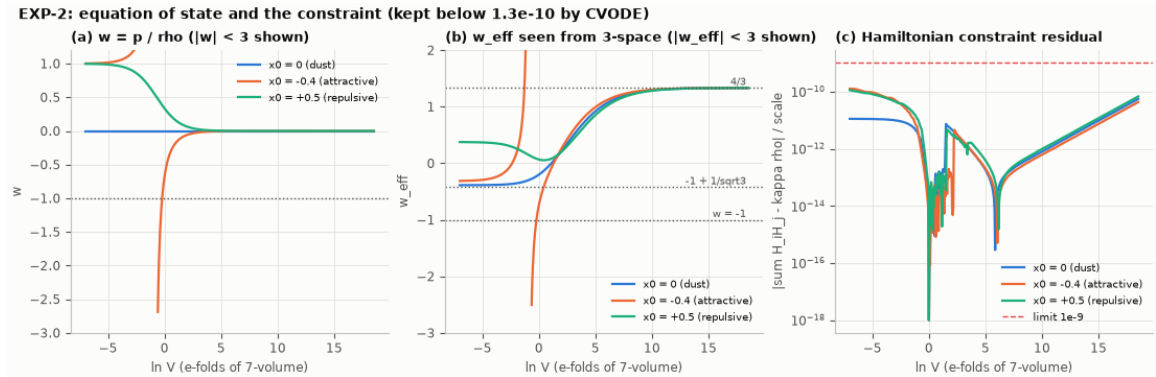


Figure 7: EXP-2 (a) $w = p/\rho$ and (b) the dilution-inferred w_{eff} against $\ln V$, with the lines $4/3$, $-1 + 1/\sqrt{3}$ and -1 , and (c) the relative Hamiltonian constraint residual with the limit 10^{-9} .

- **Equation of state.** The dust run keeps $w = 0$. The repulsive run is stiff near the singularity ($w \rightarrow 1$) and dust at late times. The attractive run is dust at late times; backward it becomes phantom ($w < -1$ with $\rho > 0$, equivalently $\text{KE}_L < 0$ and $M_{\text{eff}} < 0$) for $0.4 < V/V_0 < 0.8$ (14 output rows, measured volume range 0.407 to 0.779) and has negative energy density for $V/V_0 < 0.4$ (122 rows), down to $\rho = -1.06 \times 10^6$ at $V = e^{-7}$. The rows with $w < -1$ and $\rho > 0$ are exactly the rows with $\text{KE}_L < 0$. **These backward rows are outside the domain of the mean field.** In every one of the $14 + 122$ rows with $M_{\text{eff}} < 0$ the CSV has $s(u) = 1$ and $\text{energy_mode} = u^\dagger h u = M_{\text{eff}} < 0$: the single occupied mode is a negative-energy

eigenvector of h there (Section 4.6), so the phantom stage and the negative energy density of this run are artefacts of the single-mode mean field with fixed normal ordering, not established properties of the field.

- **The 8D energy bound.** On the constraint surface $\Theta^2 - 3H_a^2 - 2\kappa\rho = H_b^2 + 3H_c^2 \geq 0$, so wherever $\rho \geq 0$ (and $H_a, \Theta > 0$), $\Theta \geq \sqrt{3}H_a$; this gives $w_{\text{eff}} \geq -1 + (1+w)/\sqrt{3}$ only when $1+w \geq 0$, and the inequality reverses for $w < -1$. For dust the minimum measured w_{eff} is $-0.38732 > -1 + 1/\sqrt{3} = -0.42265$, and the minimum of $\Theta/(3H_a)$ over rows with $\rho > 0$ is 0.61268. In the attractive run all 14 phantom rows ($\rho > 0, w < -1$) have $w_{\text{eff}} < -1 + (1+w)/\sqrt{3}$, so there $w_{\text{eff}} < -1$ occurs with $\rho > 0$; and $\Theta < \sqrt{3}H_a$ occurs where $\rho < 0$: over all rows $\Theta/(3H_a)$ falls to 0.344, while the bound $\Theta^2 - 3H_a^2 - 2\kappa\rho$ itself stays non-negative (minimum 1.5×10^{-9}). Both situations lie in the mean-field artefact region above.

7.6 Verification

The Rust program passes 15 of 15 self-checks and the EXP-2 checker 34 of 34. Relative constraint residual at most 1.26×10^{-10} (limit 10^{-9}); $|SV/(S_0V_0) - 1|$ at most 4.03×10^{-10} ; exact volume at most 1.0×10^{-10} ; finite-difference residuals of continuity 2.1×10^{-10} , of the Einstein evolution 5.1×10^{-11} and of the Dirac equation 2.6×10^{-8} (559 rows); an Einstein tensor computed from the metric agrees to 1.1×10^{-10} . The spinor needs care: its shape error (phase invariant) is 2.0×10^{-10} , and its phase error of 2.24×10^{-7} at $t \sim 10^4$ stays below the bound $\max |M_{\text{eff}}| N_{\text{steps}} \text{ulp}(t_{\text{end}})/2 = 1.32 \times 10^{-6}$. This phase error is the rounding of CVODE's internal time $t_n += h$ with the capped step, not truncation: it does not shrink in the refined run (2.71×10^{-7}), while the shape error shrinks to 5.2×10^{-11} and the gravity error from 1.6×10^{-11} to 3.5×10^{-12} . The checker confirms the cause: the per-binade drift rates match an exact rational-arithmetic prediction to 1.7% over 21 binades. Repeat-run byte identity holds.

The Mathematica notebook derives the constraint and the evolution equations from the metric and verifies that the closed form solves and propagates them. NDSolve agrees with Rust to 3.16×10^{-8} in $\ln h_i$ and 3.24×10^{-8} in H_i/Θ . These largest deviations sit at the backward end, where the problem is ill-conditioned; divided by the amplification factor $1 + |t_s|/(t - t_s)$ (the measure of the Rust self-check) Rust's error against the closed form is 1.1×10^{-11} ; NDSolve's own error against the closed form is 2.7×10^{-11} in $\ln h_i$, so the raw deviation is on the Rust side. The spinor agrees to 1.08×10^{-7} raw and 4.2×10^{-10} after phase alignment, and the Mathematica exact-spinor calculation reproduces Rust's own maximum exact-spinor error to 8×10^{-17} .

7.7 Physical interpretation

In a self-consistent 8D cosmology the condensate's energy density falls as $1/V$ and its pressure as $1/V^2$, so every run ends as isotropic 8D dust whatever λ is. λ only matters near the singularity: repulsion makes the matter stiff, attraction drives M_{eff} through zero, after which the equations give phantom and then negative energy density, in the region where the mean field no longer describes a positive-energy state. Two conclusions matter for the dark sector. First, the extra times do not stay deflating: 8D gravity with this matter drives all seven directions to the same expansion, and a 3-space observer then sees the dust dilute like a^{-7} ($w_{\text{eff}} \rightarrow 4/3$), not like a^{-3} . Dust-like behaviour in 3-space therefore requires stabilised extra dimensions, which EXP-3 and EXP-4 assume and nothing here provides. Second, with non-negative energy density, $w \geq -1$ and $H_a, \Theta > 0$, a 3-space

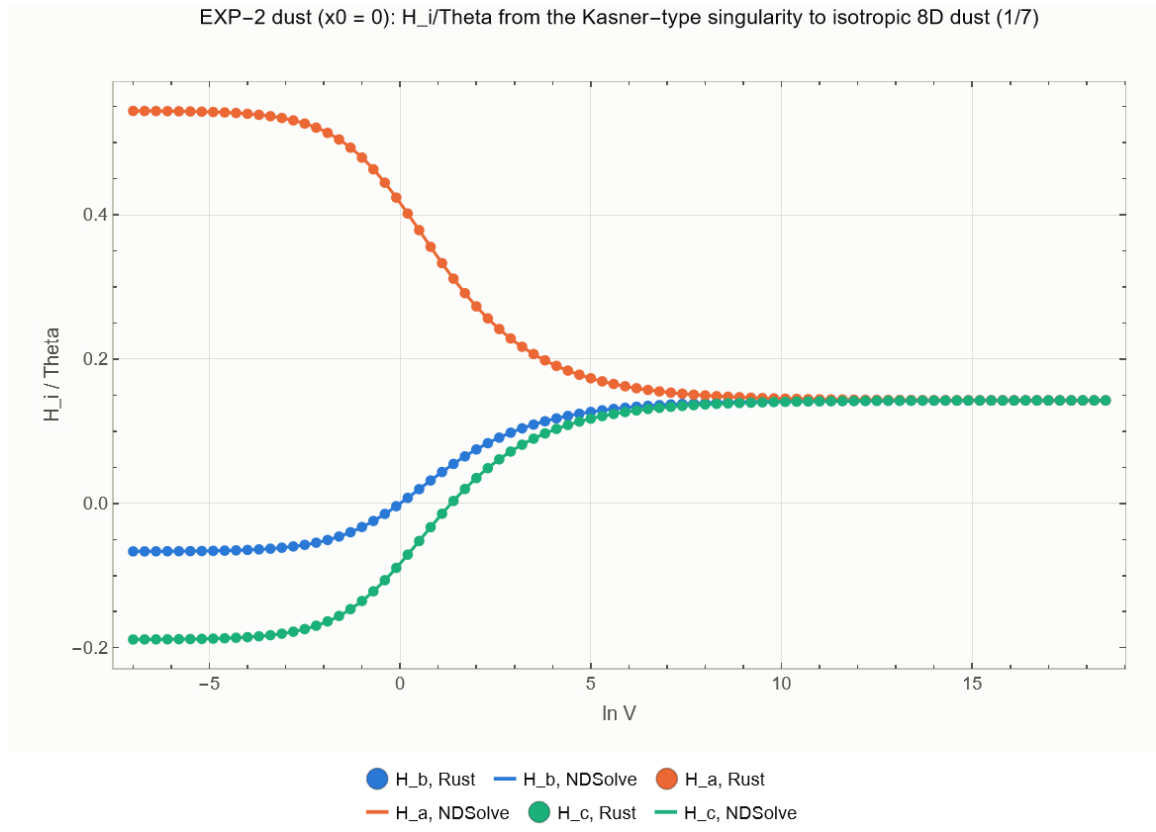


Figure 8: Mathematica cross-check of the dust run: H_i/Θ from the Kasner-type singularity to isotropic 8D dust (1/7); dots Rust, lines NDSolve.

observer can never infer $w_{\text{eff}} < -1 + (1 + w)/\sqrt{3}$; for dust that is -0.42265 . For dust, apparent dark-energy-like dilution close to $w_{\text{eff}} = -1$ needs negative 8D energy density; with w near or below -1 and $\rho > 0$ the bound does not apply.

8 EXP-3: the condensate as late-time dark energy

8.1 Purpose

EXP-3 assumes stabilised extra dimensions (b, c constant) and a 4-dimensional effective Friedmann equation, and asks whether an attractive condensate can be the dark energy described by the dark-energy fits quoted for Unite in the reference PDF.

8.2 Equations solved

With $N = \ln a$ ($a = 1$ today), $\sigma = S/S_0$ and $A = \Omega_\psi/(1 + x_0)$:

$$E^2 = \frac{H^2}{H_0^2} = \Omega_r a^{-4} + \Omega_m a^{-3} + \rho_\psi, \quad \rho_\psi = A \sigma (1 + x_0 \sigma), \quad p_\psi = A x_0 \sigma^2, \quad w = \frac{x_0 \sigma}{1 + x_0 \sigma},$$

$$\text{KE}_L = \frac{\Omega_\psi}{2} \frac{\sigma(1 + 2x_0 \sigma)}{1 + x_0}, \quad \text{PE}_L = \frac{\Omega_\psi}{2} \frac{\sigma}{1 + x_0}, \quad \text{KE}_H = 0, \quad \text{PE}_H = \rho_\psi.$$

The condensate is obtained from the evolving spinor, not assumed: $\sigma = a^{-3} s(u)/s(u_0)$. The ODE state (35 reals, independent variable N) is

$$\frac{d(H_0 t)}{dN} = \frac{1}{E}, \quad \frac{dD_C}{dN} = -\frac{1}{aE}, \quad \frac{du}{dN} = -i \frac{M_{\text{eff}}}{H_0} \frac{(-i\gamma^4)u}{E}, \quad \frac{M_{\text{eff}}}{H_0} = \mu \frac{1 + 2x_0 \sigma}{1 + 2x_0},$$

$$\frac{d \ln \sigma}{dN} = -3 + \frac{2 \text{Re}(u^\dagger (-i\gamma^4) du/dN)}{s(u)},$$

with D_C the comoving distance in units of c/H_0 and $\text{DM}(z) = 5 \log_{10}((1 + z)D_C)$ plus a constant. $\mu = M_{\text{eff}}(a = 1)/H_0 \in \{3, 7\}$ is a reduced frequency, used to show that ρ, p and w do not depend on the spinor phase (real fermions have $m/H_0 > 10^{30}$). Further diagnostics: $\rho_\psi = 0$ at $a = |x_0|^{1/3}$; $w = -1$ (where $M_{\text{eff}} = 0$) at $a = (2|x_0|)^{1/3}$; a bounce $E^2 = 0$ at the root in $(0, 1)$ of $(\Omega_m + A)a^3 + \Omega_r a^2 + A x_0 = 0$; $q_{\text{dec}} = (2\Omega_r a^{-4} + \Omega_m a^{-3} + \rho_\psi + 3p_\psi)/(2E^2)$; the adiabatic sound speed $c_s^2 = dp/d\rho = 2x_0 \sigma/(1 + 2x_0 \sigma)$; and the tangent CPL parameters $w_0 = w(1) = x_0/(1 + x_0)$, $w_a = -dw/da|_1 = 3x_0/(1 + x_0)^2$.

8.3 Initial data and parameters

$\Omega_r = 0.00009$, $\Omega_m = 0.305$, $\Omega_\psi = 0.69491$ (flat). These density parameters are inputs assumed by the numerical programme; the reference PDF gives no Ω_m , and the Ω_m of the Unite fits is not known here. $x_0 \in \{-0.462654, -0.433107, -0.3, -0.2, 0\}$: the first reproduces $w_0 = -0.861$, the second $w_0 = -0.764$, and $x_0 = 0$ is dust. $\mu \in \{3, 7\}$, so 10 runs. The initial spinor is the positive-energy rest

eigenvector of $h(a = 1)$ with $B = +1$. Each run is integrated from $N = 0$ backward (320 intervals towards $a = 1/3.5$) and forward (160 intervals to $a = 2$).

Deviation from the programme. For every $x_0 < 0$ the attractive term $\rho_\psi \propto -a^{-6}$ dominates in the past and E^2 reaches zero at a bounce $a_b > 1/3.5$. Smaller a does not exist in these models, so the backward branch stops at the last grid point with $N \geq N_b + 0.01$, and a check confirms that the stop sits exactly there. Only the dust model reaches $a = 1/3.5$.

8.4 Solver settings and statistics

Adams + fixed point, rtol 10^{-11} , atol 10^{-13} , max_step 0.01 in N : 6074 steps and 10115 right-hand-side evaluations over 20 integrations.

8.5 Results

x_0	w_0	w_a (tangent)	c_s^2 today
−0.462654	−0.861	−4.807	−12.39
−0.433107	−0.764	−4.043	−6.47
−0.3	−0.4286	−1.837	−1.50
−0.2	−0.25	−0.9375	−0.667
0	0	0	0

The characteristic epochs, as redshift z and scale factor a :

x_0	$\rho_\psi = 0$	$w = -1$	bounce $E^2 = 0$
−0.462654	$z = 0.293, a = 0.773$	$z = 0.0262, a = 0.974$	$z = 0.388, a = 0.721$
−0.433107	$z = 0.322, a = 0.757$	$z = 0.0490, a = 0.953$	$z = 0.423, a = 0.703$
−0.3	$z = 0.494, a = 0.669$	$z = 0.186, a = 0.843$	$z = 0.633, a = 0.612$
−0.2	$z = 0.710, a = 0.585$	$z = 0.357, a = 0.737$	$z = 0.890, a = 0.529$
0	none	none	none

The deceleration and the bare masses of the two runs per coupling:

x_0	q_{dec} today	$q_{\text{dec}} = 0$ at	m/H_0 ($\mu = 3, 7$)
−0.462654	−0.397	$z = -0.126, a = 1.144$	40.2, 93.7
−0.433107	−0.296	$z = -0.103, a = 1.115$	22.4, 52.3
−0.3	0.0533	$z = 0.0290, a = 0.972$	7.5, 17.5
−0.2	0.239	$z = 0.191, a = 0.840$	5.0, 11.7
0	0.500	none	3, 7

- **The changing equation of state.** For $x_0 < 0$, w rises with time ($w_a < 0$, the thawing sign, but not a thawing field): in the equations it comes up from $-\infty$ at the pole where $\rho_\psi = 0$,

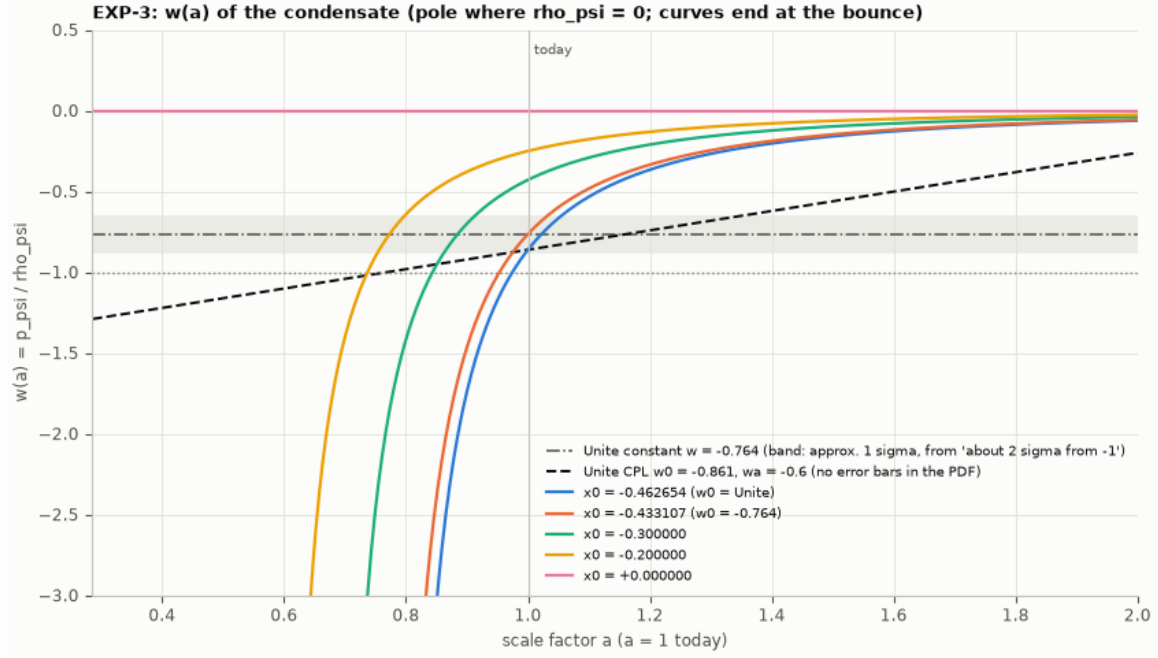


Figure 9: EXP-3 $w(a) = p_\psi/\rho_\psi$ of the condensate for the five couplings, with the Unite CPL line $w_0 = -0.861$, $w_a = -0.60$ and the Unite constant $w = -0.764$. The shaded band around -0.764 (± 0.118) is only an approximate one-sigma band inferred from the PDF's statement "roughly two standard deviations from -1 "; the PDF gives no error bars for (w_0, w_a) , so none is drawn for the CPL line. The curves have a pole where $\rho_\psi = 0$ and end at the bounce.

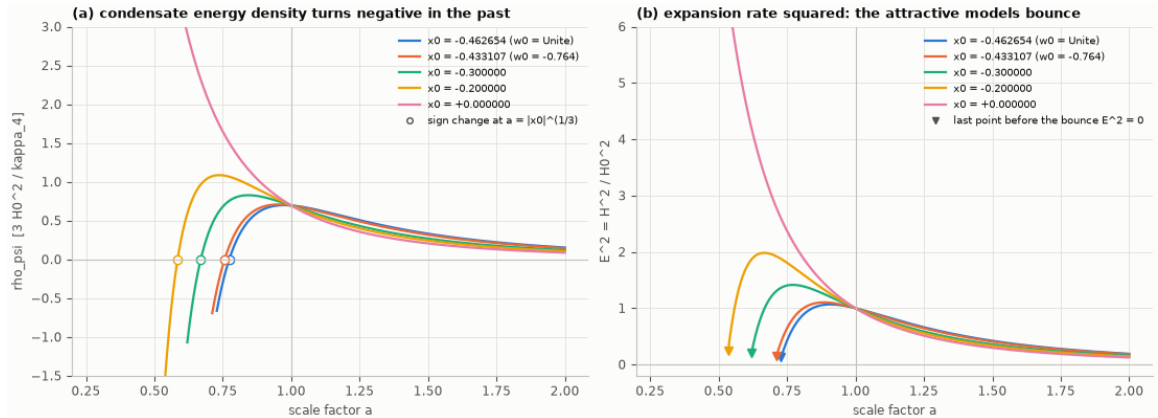


Figure 10: EXP-3 (a) the condensate energy density $\rho_\psi(a)$, which changes sign at $a = |x_0|^{1/3}$ (circles), and (b) $E^2 = H^2/H_0^2$, which reaches zero at the bounce for every attractive model (triangles: last point before the bounce).

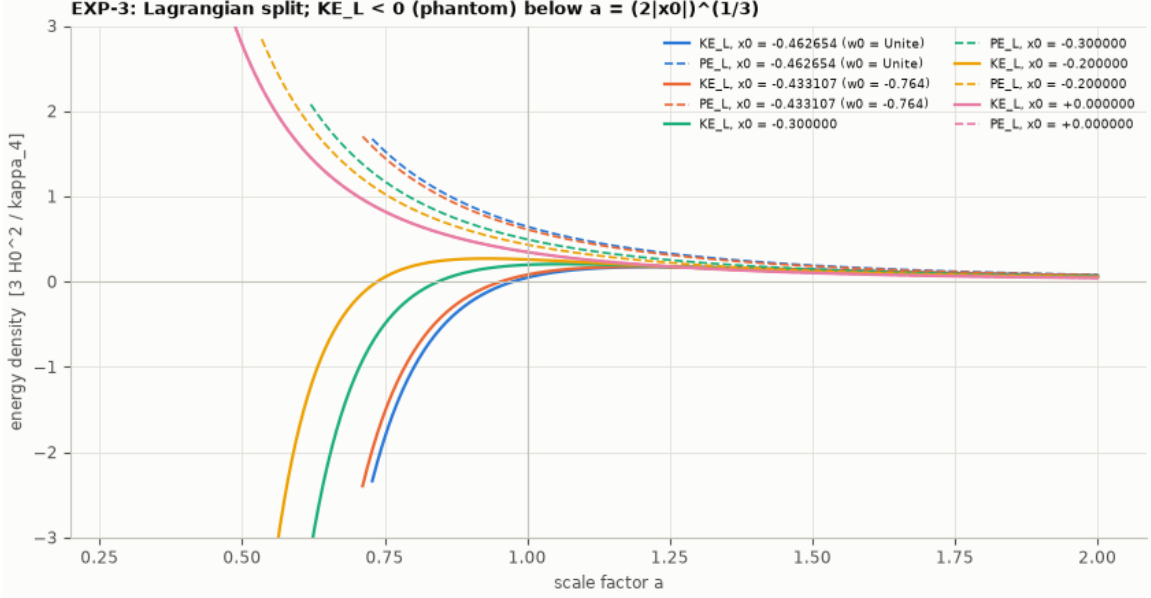


Figure 11: EXP-3 Lagrangian split: KE_L (solid) turns negative, and w drops below -1 , for $a < (2|x_0|)^{1/3}$, where the mean-field mode is a negative-energy level (Section 4.6); PE_L (dashed) stays positive.

crosses -1 where $M_{\text{eff}} = 0$ ($KE_L = 0$), passes w_0 today and tends to 0 in the future, because $p_\psi \propto a^{-6}$ dies away faster than ρ_ψ . Energy density and pressure change for one reason only: $\sigma = a^{-3}$.

- **Validity of the mean field.** Before the crossing ($a < (2|x_0|)^{1/3}$) $M_{\text{eff}} < 0$ while $s(u) = 1$: the occupied mode is a negative-energy level of h (Section 4.6). In the committed CSV files of every attractive run all phantom rows and all rows with $\rho_\psi < 0$ lie there, for example all rows with $a \leq 0.973$ for $x_0 = -0.462654$. The phantom epoch, the negative energy density and the bounce are therefore artefacts of the mean field; the Unite-tuned model leaves the valid regime at $z = 0.026$.
- **Acceleration.** The two Unite-tuned models accelerate today ($q_{\text{dec}} = -0.397$ and -0.296) and all the way back to the bounce, and they start to decelerate again at $a = 1.144$ and 1.115 when the condensate becomes dust-like. Neither has a matter-dominated era in these equations, and before $z = 0.026$ and $z = 0.049$ neither is inside the validity of the mean field. The weaker couplings decelerate today and accelerate only in the past, near the bounce.
- **Phase independence.** $\mu = 3$ and $\mu = 7$ give the same ρ (1.74×10^{-9}), p (3.00×10^{-9}) and w (2.77×10^{-9} , condition-scaled): only $s(u) = 1$ matters, not the phase.
- **Sound speed.** $c_s^2 = dp/d\rho$ is negative today for every $x_0 < 0$ (-12.39 for the Unite-tuned model). In a fluid description that is a gradient instability of perturbations; perturbations themselves were not computed.

The fits and the comparison with Unite are tabulated in Section 12.4.

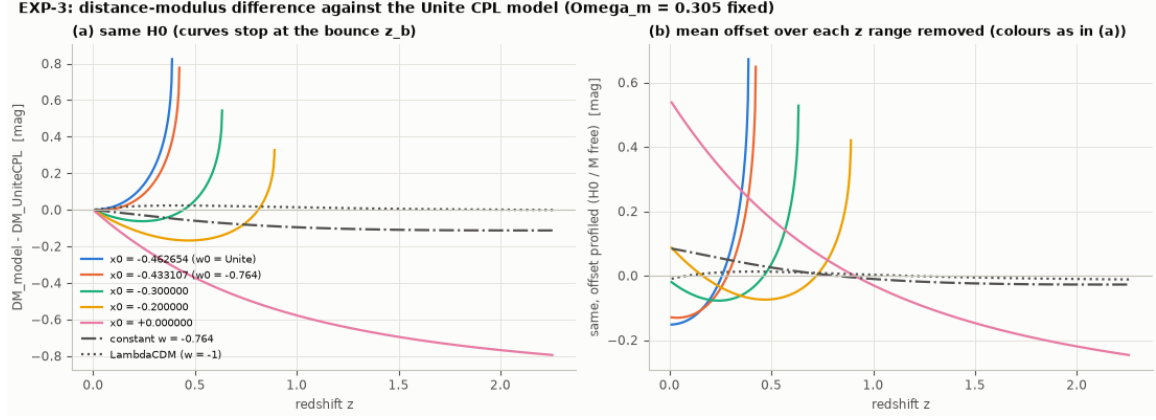


Figure 12: EXP-3 distance modulus minus that of the Unite CPL model ($\Omega_m = 0.305$ fixed) for the five couplings, a constant $w = -0.764$ and Λ CDM: (a) same H_0 , curves ending at the bounce; (b) with the mean offset over each curve's redshift range removed (H_0 and absolute magnitude free).

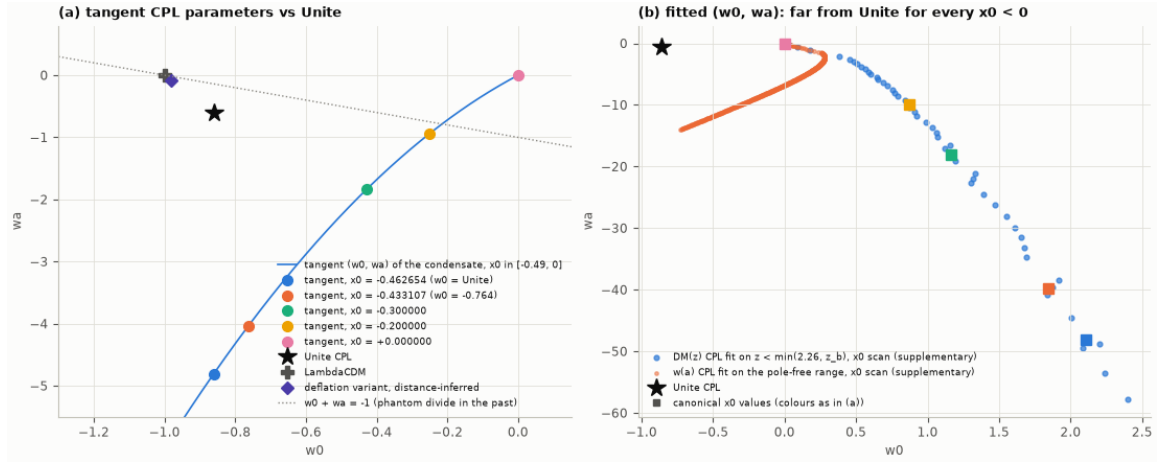


Figure 13: EXP-3 (w_0, w_a) plane. (a) the tangent CPL parameters of the condensate for $x_0 \in [-0.49, 0]$ with the canonical couplings, the Unite point, Λ CDM, the line $w_0 + w_a = -1$ and the distance-inferred point of the deflation variant; (b) the supplementary restricted fits of the fine scan, far from Unite for every $x_0 < 0$.

8.6 Verification

The Rust program passes 14 of 14 self-checks, the EXP-3 checker 31 of 31 and the analysis 10 of 10 (Nelder-Mead recovers synthetic CPL and constant- w curves to 2.0×10^{-12} and 1.4×10^{-12} , and a synthetic wCDM curve with Ω_m free to 4.6×10^{-13} ; Rust distances match an independent quadrature to 1.7×10^{-9}). σ from the spinor agrees with a^{-3} to 2.50×10^{-9} (refined: 2.10×10^{-10}) and $\exp(\ln \sigma)$ to 1.95×10^{-14} . The time and distance agree with independent quadratures to 1.11×10^{-9} and 1.75×10^{-9} , the spinor with its exact phase to 5.48×10^{-8} (refined 6.41×10^{-9}). The roots are reproduced to 2.6×10^{-9} ($\rho_\psi = 0$), 2.3×10^{-9} ($w = -1$) and 1.5×10^{-9} ($q_{\text{dec}} = 0$), the tangent w_a to 2.8×10^{-7} relative, and the bounce stop to 1.5×10^{-15} . The Mathematica notebook derives the Friedmann constraint, the continuity equation, w and q_{dec} symbolically; NDSolve agrees to 2.12×10^{-8} in the spinor (1.25×10^{-9} phase aligned) and to 1.95×10^{-14} in $\ln \sigma$, and its exact spinor error is 2.0×10^{-12} . The finite-difference negative control with a flipped mass sign fails as it should (2.0). The 4D-effective Friedmann equation with stabilised extra dimensions is an assumption of this experiment, not a consequence of the 8D equations; the Mathematica notebook states this too.

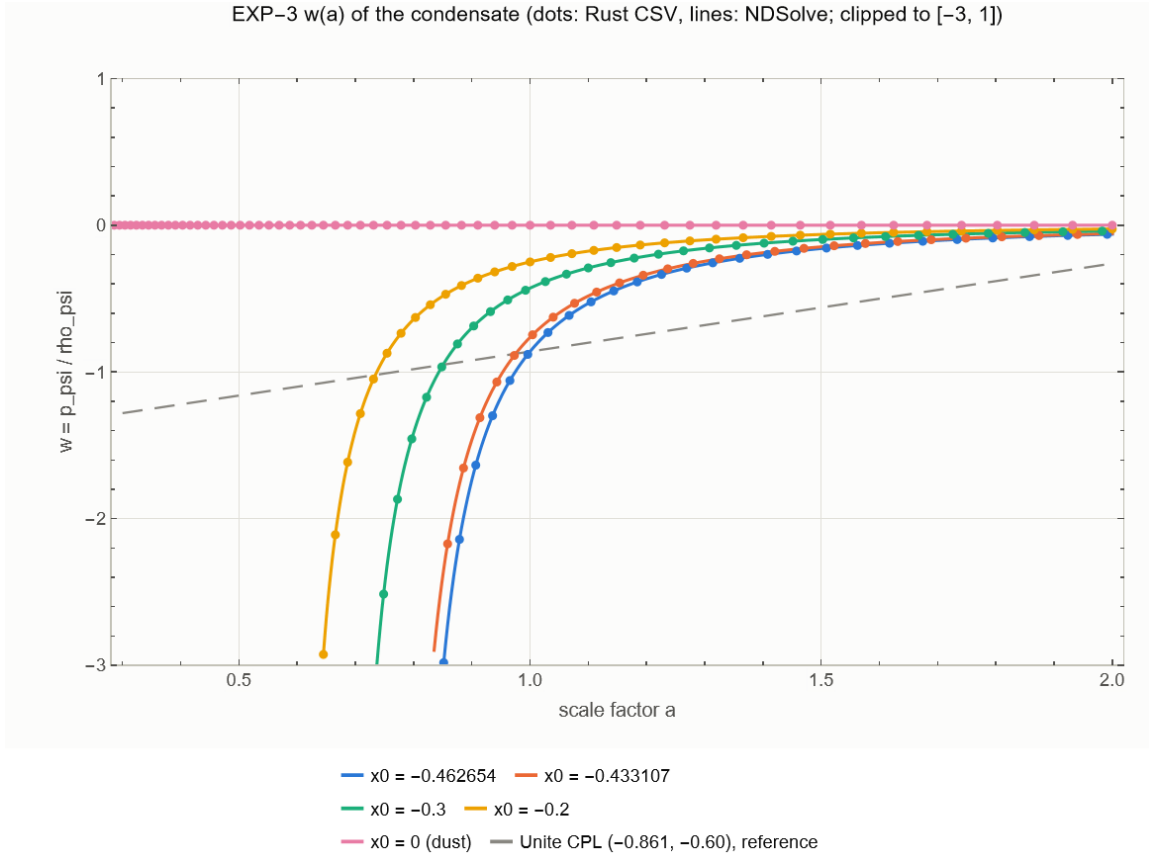


Figure 14: Mathematica cross-check of $w(a)$: dots Rust CSV, lines NDSolve, with the Unite CPL line for reference; clipped to $[-3, 1]$.

8.7 Physical interpretation

The attractive condensate is the only object in this framework whose own pressure is negative. Its negative pressure is $\frac{\lambda}{2}S^2 \propto a^{-6}$, so it matters only at high density, that is in the past, and there it wins over the positive mS : going back in time the effective mass passes through zero, and the equations then continue to negative energy density and to $H = 0$. Today's w_0 can be tuned to any value in $(-1, 0)$ by the single parameter x_0 , but $w_a = 3x_0/(1+x_0)^2$ is then fixed, and the evolution is fast because the negative-pressure part dilutes like a^{-6} . The crossing of $w = -1$ is not the smooth crossing of a non-canonical scalar and not an established phantom crossing: it is the instant where the effective mass vanishes, and on the far side of it the single mean-field mode occupies a negative-energy level. Everything computed there (phantom w , $\text{KE}_L < 0$, $\rho_\psi < 0$, the bounce) is a mean-field artefact of fixed normal ordering.

9 EXP-4: dirac16complex quanta as dark matter

9.1 Purpose

EXP-4 assumes a 3-space FRW universe with the hidden space and the extra times static ($b = c = 1$) and asks two things: how the equation of state of a thermal gas of quanta changes from relativistic to non-relativistic, and how many quanta the transition from inflation to radiation creates.

9.2 Equations solved

A mode with comoving momentum k along x_1 and physical momentum $K = k/a$ obeys

$$i\dot{u} = h(t)u, \quad h = -im\gamma^4 - K\gamma^4\gamma^1, \quad h^2 = E^2, \quad E = \sqrt{m^2 + K^2}, \quad \dot{K} = -KH.$$

Only 3-space momenta are kept: the hidden-space momentum k_0 is set to zero. The hidden direction x_0 is space-like, belongs to the good sector and is static here ($b = 1$), so this is an assumption, not a consequence: it holds for a compact hidden dimension whose Kaluza-Klein gap is much larger than T and H_{inf} , or by hand. A thermal gas at $T = 10m$ would otherwise populate k_0 ; the relativistic 3-space pressure would then be $\rho/4$, not $\rho/3$, and the density of states would change. Per mode $\varepsilon = u^\dagger h u$, $p_1 = -K u^\dagger \gamma^4 \gamma^1 u$, $s = u^\dagger (-i\gamma^4) u$, with the identity $\varepsilon = ms + p_1$; $|\beta_k|^2 = |P_- u|^2 / |u|^2$ is the weight on the instantaneous negative-energy subspace, and the first-order adiabatic $|\beta_k|^2$ removes from it the adiabatic dressing $(mKH/(4E^3))^2$ of an unexcited mode. Structure used for the counting and verified numerically: $h = (m\sigma_z + K\sigma_x) \otimes I_8$ with $\sigma_z = -i\gamma^4$ and $\sigma_x = -\gamma^4\gamma^1$, so there are 8 spin states per k , and $Ch^*C = -h$ maps particles to antiparticles.

(a) Thermal gas. $a(t) = (t/t_i)^{1/2}$ with $t_i = 1/(2H_i)$, and with $W_n = \frac{16}{2\pi^2} w_n k_n^2 f_n$, $f = 1/(e^{E_i/T_i} + 1)$ and 16 states per k (8 particles and 8 antiparticles):

$$\rho = \frac{1}{a^3} \sum_n W_n \varepsilon_n, \quad p = \frac{1}{3a^3} \sum_n W_n p_{1,n}, \quad \text{KE}_H = 3p, \quad \text{PE}_H = \rho - 3p,$$

compared with the kinetic-theory integrals

$$\rho_{\text{kin}} = \frac{16}{2\pi^2 a^3} \int k^2 f E dk, \quad p_{\text{kin}} = \frac{16}{2\pi^2 a^3} \int k^2 f \frac{K^2}{3E} dk.$$

(b) Pair creation. $a = e^{H_{\text{inf}} t}$ for $t < 0$, glued C^1 at $t = 0$ to $a = (1 + 2H_{\text{inf}} t)^{1/2}$: a and H are continuous and $\dot{H}$ jumps from 0 to $-2H_{\text{inf}}^2$. The integration is restarted at $t = 0$. Each mode starts in the first-order adiabatic vacuum at $k/a(t_0) = 200H_{\text{inf}}$. With 8 filled negative-energy states per k in the Dirac sea, each ending with weight $|\beta_k|^2$ on the positive-energy subspace, and with $|\beta_k|^2$ the final value in the first-order adiabatic basis (the instantaneous-basis final value is dominated for $k \gtrsim 12$ by the adiabatic dressing at the end of the run, oscillates in k and depends on where the run stops; its integrals are kept only as the literal definition),

$$na^3 = \frac{16}{2\pi^2} \int k^2 |\beta_k|^2 dk, \quad \rho a^3 = \frac{16}{2\pi^2} \int k^2 |\beta_k|^2 E_k(a) dk, \quad pa^3 = \frac{16}{2\pi^2} \int k^2 |\beta_k|^2 \frac{K^2}{3E} dk,$$

counting particles and antiparticles. For large k the kink at $t = 0$ predicts $|\beta_k|^2 \rightarrow (mk/(4E^4))^2$ at $a = 1$. The grid stops at $k = 40$; the analytic kink tail beyond it is integrated separately (64-node Gauss-Legendre in k_{max}/k) and reported as a correction to na^3 , ρa^3 and pa^3 .

9.3 Initial data and parameters

(a) $m = 1$, comoving temperature $T_i = 10$ at $a = 1$, $H_i = 0.05$, $t_i = 10$, a from 1 to 100 ($t_{\text{end}} = 10^5$), 60 output intervals uniform in $\ln a$; 48 Gauss-Legendre nodes on $[0, k_{\text{max}}]$, $k_{\text{max}} = 12T_i = 120$. Each mode starts from the positive-energy $B = +1$ eigenvector of $h(t_i)$. Extras: a second spin state for four nodes, a negative-energy mode for one node, and the first-order adiabatic vacuum for two trans-relativistic nodes. **(b)** $H_{\text{inf}} = 1$, $m/H_{\text{inf}} \in \{0, 0.1, 0.5, 1, 2\}$, 64 Gauss-Legendre nodes in $\ln k$ on $[10^{-3}, 40]$; the run ends when $H = 10^{-4}m$ ($a_{\text{end}}^2 = 10^4 H_{\text{inf}}/m$; for $m = 0$ the $m = 0.1$ value); antiparticle runs for three nodes at $m = 1$.

9.4 Solver settings and statistics

Adams + fixed point, rtol 10^{-13} , atol 10^{-14} , max_step 2.0; 287898778 steps and 435564940 right-hand-side evaluations. Adams was selected by a recorded test at the default tolerances against a reference run at rtol/10: for node 2 ($k = 0.9525$) Adams had error 1.25×10^{-12} with 1370 evaluations against BDF's 2.27×10^{-10} with 4730 (plus 2464 for the Jacobian); for node 47 ($k = 119.93$) 1.71×10^{-9} with 88979 against 1.84×10^{-8} with 313247 (plus 164384). The tolerance is tight because each thermal mode turns through about 10^5 radians: $\max |u^\dagger u - 1|$ is 2.4×10^{-4} at rtol 10^{-10} and 2.4×10^{-7} at 10^{-13} .

9.5 Results

- **Relativistic to dust.** $w = 0.3328543348$ at $a = 1$ (kinetic theory: the same to ten digits) and $w = 0.0359224$ at $a = 100$ (kinetic theory 0.0359223), both on the truncated grid $k \leq k_{\text{max}} = 12T_i$. ρa^4 is constant early and ρa^3 late. The momentum energy $\text{KE}_H = 3p$ gives way to rest-mass energy.

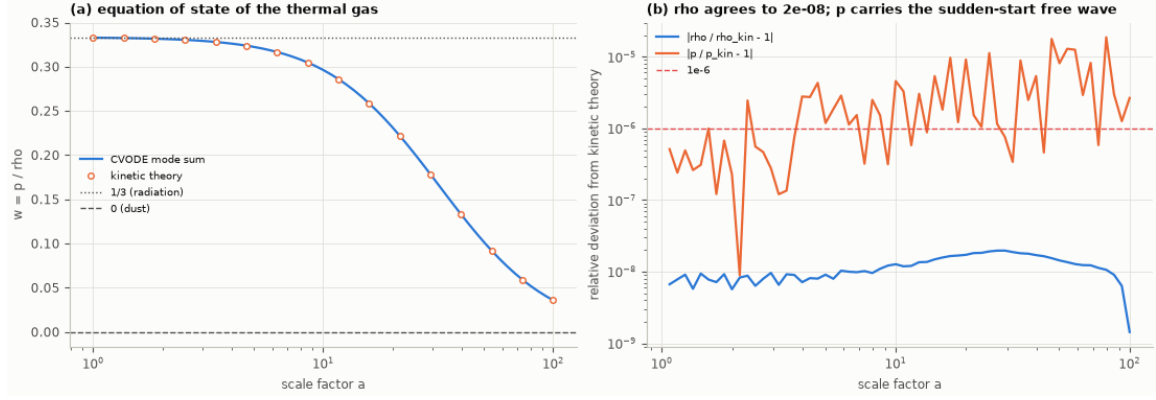


Figure 15: EXP-4 thermal gas. (a) $w = p/\rho$ of the CVODE mode sum and of kinetic theory against a , from $1/3$ towards 0; (b) relative deviations of ρ (about 10^{-8}) and of p (up to 1.9×10^{-5} , the sudden-start free wave) from kinetic theory.

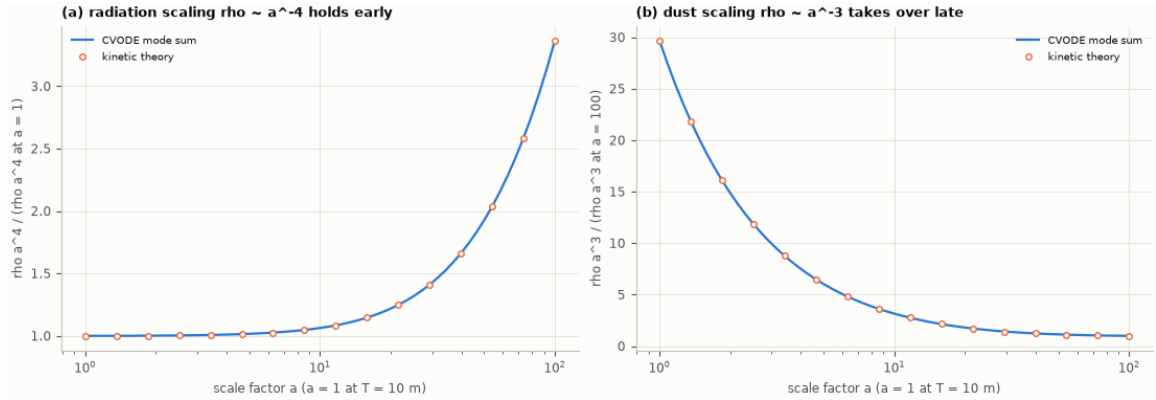


Figure 16: EXP-4 thermal gas scaling: (a) ρa^4 , constant while the gas is relativistic, and (b) ρa^3 , constant once it is non-relativistic.

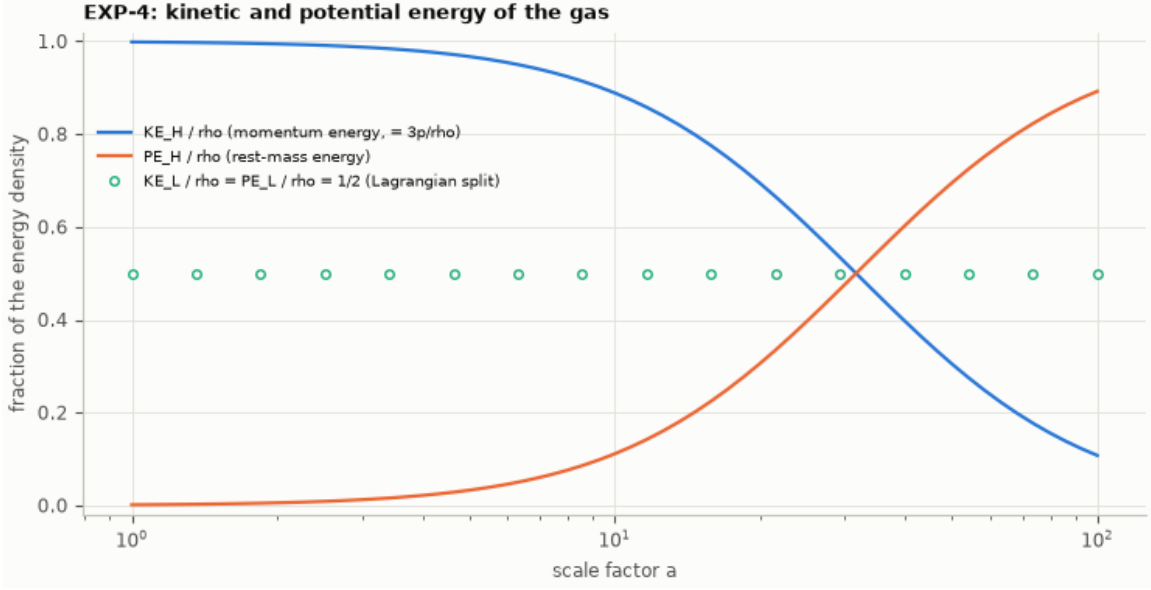


Figure 17: EXP-4 kinetic and potential energy of the gas: $KE_H/\rho = 3w$ (momentum energy) falls from 1 and PE_H/ρ (rest mass) rises towards 1, while the Lagrangian split stays at $KE_L/\rho = PE_L/\rho = 1/2$.

- **Agreement with kinetic theory.** The mode sum and the kinetic comparison share the cut $k \leq k_{\max}$, so this comparison checks the ODE integration, not the truncation. ρ agrees to 1.98×10^{-8} . p deviates by up to 1.87×10^{-5} , independent of the tolerance: the start from the instantaneous eigenvector leaves a free negative-frequency wave that enters p_1 and s at first order, $p_1 = (1 - 2|\beta|^2)K^2/E + 2\text{Re}(\alpha^*\beta)mK/E$, while ε has no such term. The deviation stays inside the independently predicted interference envelope (ratio 0.28), and the adiabatic-vacuum runs reduce the single-mode pressure deviation from 1.78 to 0.12 relative.
- **No spurious particle production in the gas.** The occupation-weighted $|\beta|^2$ is 3.7×10^{-8} and the adiabatic-vacuum runs end at 5.6×10^{-8} . The largest single-mode value, 6.37×10^{-5} at $k = 0.953$, is 0.998 times $4|c|^2$ with $|c|^2 = (mkH_i/(4E_i^3))^2$, the sudden-start wave, not particle production.
- Spin independence and particle-antiparticle symmetry are exact (deviation 0.0).
- **Momentum truncation.** Truncating the momenta at $k_{\max}$ misses 2.4×10^{-3} of ρ at $a = 1$ and, at $a = 100$, 9.0×10^{-4} of ρ and 5.3×10^{-3} of p (the checker's kinetic theory on $[0, 60T_i]$). Without the cut the Fermi-Dirac gas has $w = 0.0361$ at $a = 100$ instead of 0.0359; the quoted $w(100)$ therefore refers to the truncated grid. The truncation is reported, not corrected.

The produced number on the grid $k \leq 40$, from the final spectrum in the first-order adiabatic basis:

m/H_{inf}	na^3 in H_{inf}^3	$\max \beta_k ^2$	a_{end}^2
0	1.8×10^{-24}	8.5×10^{-25}	10^5
0.1	1.454×10^{-3}	0.348	10^5
0.5	4.990×10^{-3}	0.0640	2×10^4

m/H_{inf}	na^3 in H_{inf}^3	$\max \beta_k ^2$	a_{end}^2
1	4.412×10^{-3}	0.0138	10^4
2	2.421×10^{-3}	0.00265	5×10^3

The equation of state of the produced gas, from the same final (frozen) spectrum; " w at $a = 1$ " is that spectrum evaluated at $a = 1$, a hypothetical, because the production is not finished at $a = 1$ (see below):

m/H_{inf}	w at $a = 1$	w at the end	ρa^3 at the end
0	1/3	1/3	1.3×10^{-26}
0.1	0.3081	1.20×10^{-4}	1.454×10^{-4}
0.5	0.2539	1.14×10^{-4}	2.495×10^{-3}
1	0.2386	1.62×10^{-4}	4.413×10^{-3}
2	0.2395	3.02×10^{-4}	4.845×10^{-3}

Basis and truncation. The tables use the first-order adiabatic basis. The instantaneous-basis final $|\beta_k|^2$ (the literal definition) contains, for $k \gtrsim 12$, the adiabatic dressing $(mKH/(4E^3))^2$ at the end of the run; it gives $na^3 = 1.455 \times 10^{-3}$, 4.991×10^{-3} , 4.414×10^{-3} and 2.422×10^{-3} , and at the end $w = 3.09 \times 10^{-4}$, 1.28×10^{-4} , 1.70×10^{-4} and 3.09×10^{-4} for $m = 0.1, 0.5, 1$ and 2 , values that depend on the k -grid and on where the run stops. The analytic kink tail beyond $k = 40$ adds 1.05×10^{-6} to na^3 for $m = 2$ (a fraction 4.3×10^{-4}), and 6.4×10^{-3} of ρ and 8.9×10^{-3} of p at $a = 1$; with it, $na^3 = 2.422 \times 10^{-3}$ and w at $a = 1$ is 0.2401 for $m = 2$ (0.3081, 0.2540 and 0.2388 for the lighter masses), and w at the end is 3.28×10^{-4} for $m = 2$.

- **Production for $m > 0$ only.** For $m = 0$ the mode equation is conformally invariant, $h = K(t)\sigma_x \otimes I_8$ has time-independent eigenvectors, and the measured $|\beta|^2$ (8.5×10^{-25}) is roundoff. For $m > 0$ the yield na^3 is a few $10^{-3}H_{\text{inf}}^3$, largest near $m = 0.5H_{\text{inf}}$; the maximum $|\beta_k|^2 \leq 1$ respects the Pauli principle. The high- k tail follows the kink prediction to 4.2% (limit 35%).
- **When the quanta are made.** For $m \geq 0.5H_{\text{inf}}$ most of the final number is produced after the de Sitter to radiation transition, in the radiation era when H is of order m : the adiabatic-basis na^3 at $a = 1$ in pair_history.csv is about a third to a half of its final value for $m = 0.5, 1$ and 2 (for $m = 0.1$ it is still evolving at $a = 1$).
- **The produced gas becomes dust:** the final spectrum has w between 0.24 and 0.31 when evaluated at $a = 1$ and about 10^{-4} at the end, and ρa^3 approaches mna^3 (for $m = 1$: 4.413×10^{-3} against $na^3 = 4.412 \times 10^{-3}$).

9.6 Verification

The Rust program passes 23 of 23 self-checks and the EXP-4 checker 51 of 51. The checker recomputes the grids, the initial states, every derived column and every mode sum from the raw spinors (to 2.5×10^{-15}), re-derives kinetic theory with an independent Gauss-Legendre quadrature (quadrature error 1.2×10^{-11} in ρ), checks unitarity (2.4×10^{-7} thermal, 4.1×10^{-8} pair) and

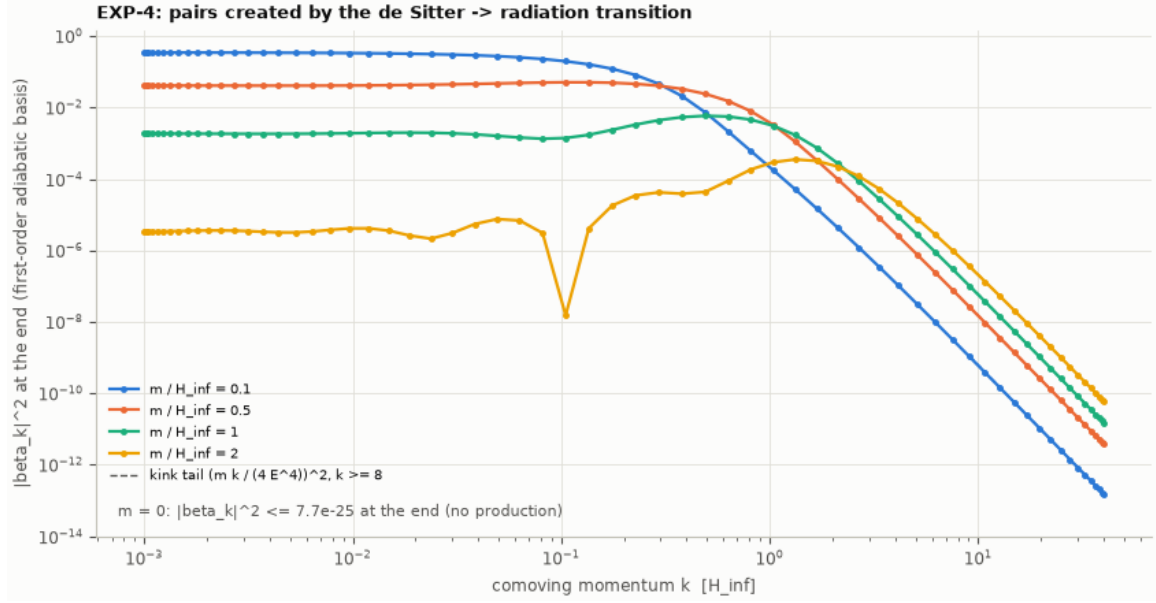


Figure 18: EXP-4 pair creation by the de Sitter to radiation transition: $|\beta_k|^2$ at the end (first-order adiabatic basis) for $m/H_{\text{inf}} = 0.1, 0.5, 1$ and 2 , with the kink tail $(mk/(4E^4))^2$ for $k \geq 8$; for $m = 0$, $|\beta_k|^2 \leq 7.7 \times 10^{-25}$ (no production).

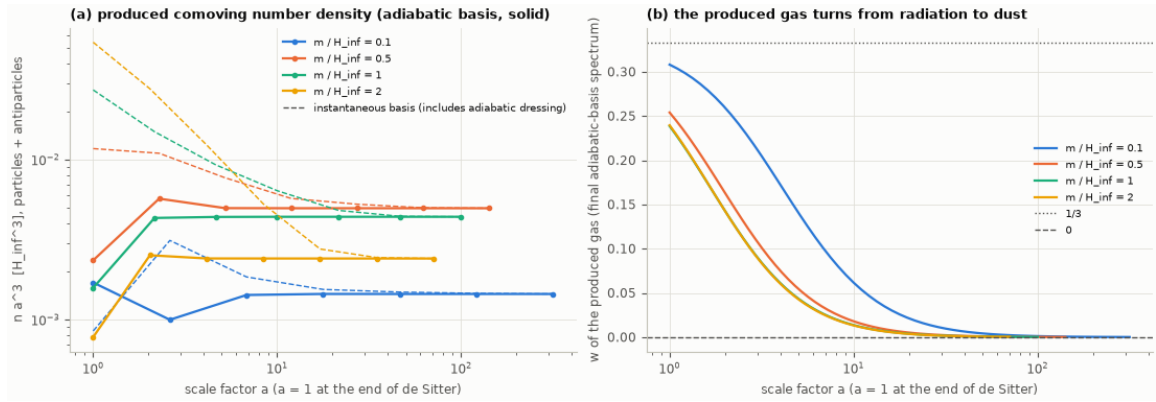


Figure 19: EXP-4 (a) the produced comoving number density na^3 (adiabatic basis solid, instantaneous basis dashed), and (b) the equation of state of the produced gas with its final adiabatic-basis spectrum, from nearly $1/3$ when evaluated at $a = 1$ to dust.

the Krein norm, integrates selected modes with an independent RK4 with Richardson estimate (1.74×10^{-9} thermal, 6.2×10^{-10} pair) because the CSV sampling is too sparse for finite differences, and checks repeat-run byte identity and refined convergence (thermal ρ and p change by 3.0×10^{-8} and 3.7×10^{-8} relative, unitarity improves from 2.4×10^{-7} to 4.4×10^{-8} , pair $|\beta|^2$ changes by 1.8×10^{-9} relative). The RK4 reference covers the thermal modes only up to $a = 3$. Over the full range $a \in [1, 100]$ the checker compares five thermal nodes (0, 2, 16, 30, 47) with a fourth-order Magnus integration of the exact two-level reduction $h = (m\sigma_z + K\sigma_x) \otimes I_8$, through the overlap $\langle u(t_i)|u(t) \rangle$, which carries the phase (Magnus self-convergence 5.9×10^{-9}). The Rust thermal states carry a phase error of at most 6.5×10^{-6} rad at $a = 100$ (maximum overlap deviation 6.0×10^{-6}), after about 10^5 radians or more of phase, and the refined run does not reduce it (maximum phase error 7.1×10^{-6} rad, overlap deviation 7.8×10^{-6} ; canonical and refined states differ by up to 1.24×10^{-5}). So the refined convergence is no evidence for the phase; the error is not a truncation error, consistent with the time-accumulation rounding identified in EXP-2, and only phase-invariant quantities (ε , p_1 , s , $|\beta|^2$) enter the results. The Mathematica notebook derives the mode equation symbolically, re-integrates four thermal nodes and ten pair modes with NDSolve and recomputes kinetic theory with NIntegrate. Its thermal nodes 0 and 2 run to $a = 100$, the two hardest nodes 16 ($k = 32$) and 47 ($k = 120$) only to $a = 10$ (a small fraction of their phase), so for those two the cross-check is partial. The results (thermal: 2.62×10^{-6} raw, 5.9×10^{-8} phase aligned; NDSolve Adams against order-8 Runge-Kutta on node 2: 6.6×10^{-9} , so the raw deviation is on the Rust side; pair: 7.6×10^{-9} phase aligned and 1.9×10^{-9} relative in $|\beta|^2$, massless $|\beta|^2 = 0$; kinetic theory by NIntegrate (ρ : 1.98×10^{-8} , p : 1.87×10^{-5} , the free-wave effect above)).

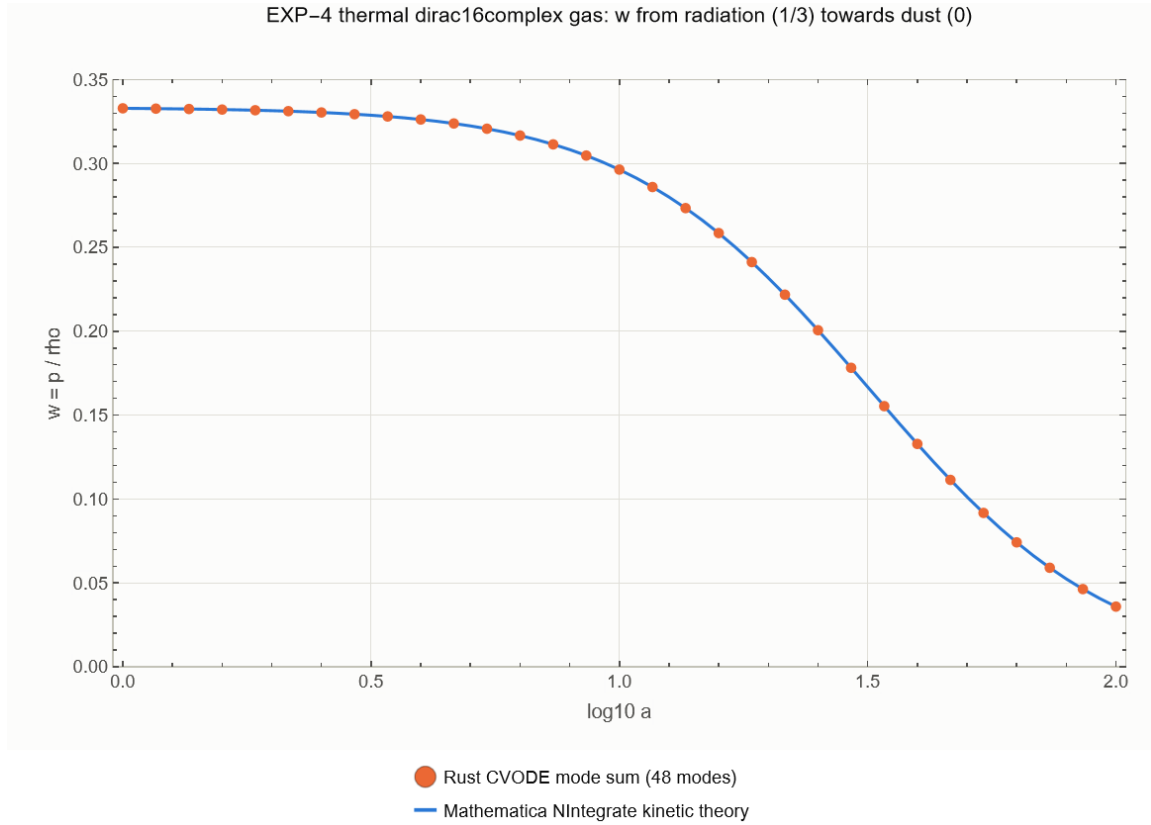


Figure 20: Mathematica cross-check of the thermal gas: w of the 48-mode Rust sum (dots) against NIntegrate kinetic theory (line).

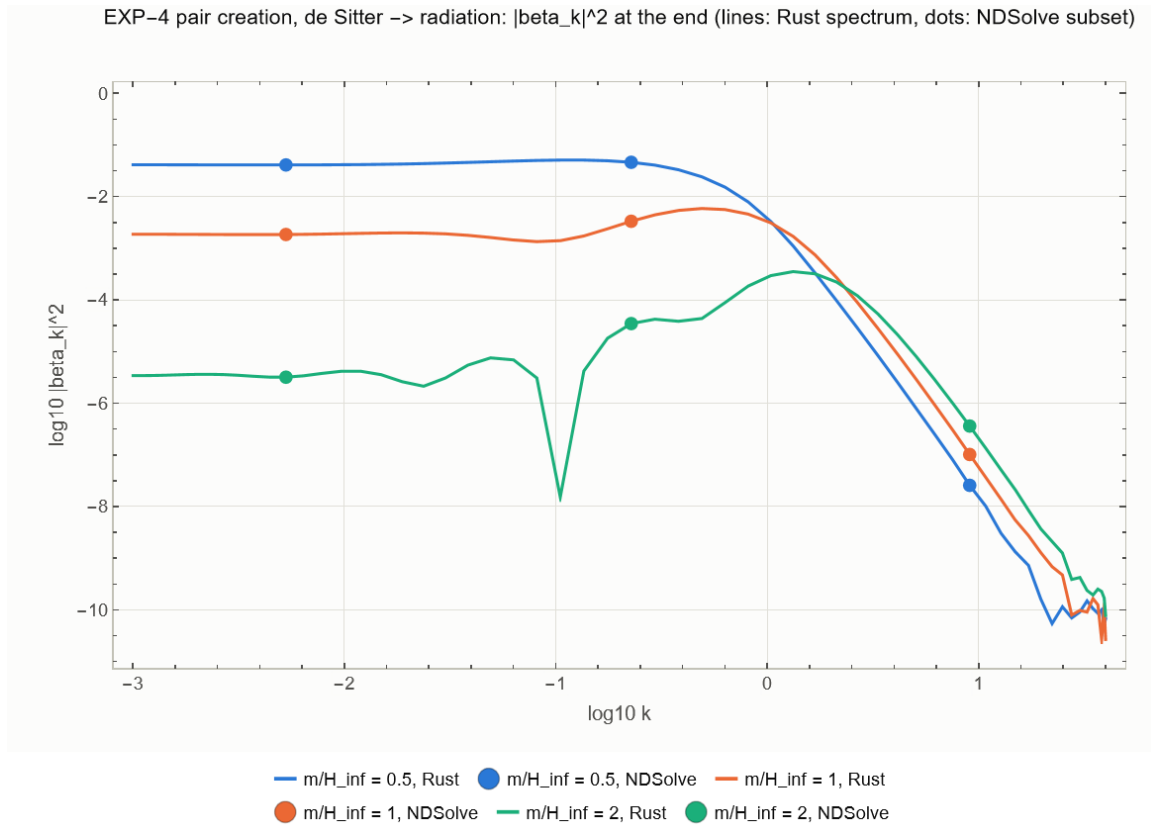


Figure 21: Mathematica cross-check of the pair spectra for $m/H_{\text{inf}} = 0.5, 1$ and 2 : lines Rust, dots NDSolve subset.

9.7 Physical interpretation

In the good sector, with the extra dimensions held static and no hidden-space momentum, `dirac16complex` quanta are an ordinary gas of massive fermions with 16 states per momentum. Their pressure is momentum flux and redshifts away, so the equation of state changes continuously from radiation to dust: the gas is hot, then warm, then cold. The expansion itself creates such quanta for any $m > 0$, mainly after the de Sitter to radiation transition, when H is of order m , in amounts of order $10^{-3}H_{\text{inf}}^3$ per comoving volume (normalised to $a = 1$ at the end of de Sitter), and the created gas is dust soon after. This is the behaviour a gravitationally produced, heavy, weakly coupled dark-matter candidate needs. What it does not give is a relic density in physical units: that would require H_{inf} and m in physical units and the post-inflationary history, none of which the framework fixes.

10 EXP-5: the extra-time instability

10.1 Purpose

Every result above uses the sector without extra-time momentum. EXP-5 shows why: a mode with momentum along an extra time is unstable once the extra times have deflated enough.

10.2 Equations solved

The extra times deflate as $c = h_5 = h_6 = h_7 = e^{-Ht}$ ($H = 1$), and a mode has momentum q along x_5 ($k = 0$, $m = 1$), so $k_5/h_5 = Q(t) = qe^{Ht}$ and

$$h(t) = -im\gamma^4 - Q(t)\gamma^4\gamma^5, \quad h^2 = E^2(t) = m^2 - q^2e^{2Ht}, \quad t_* = \frac{\ln(m/q)}{H}.$$

h is not Hermitian ($\gamma^4\gamma^5$ is real antisymmetric) but it is B -pseudo-Hermitian, $h^\dagger B = Bh$: the Krein norm $u^\dagger Bu$ is conserved exactly and the Hilbert norm $u^\dagger u$ is not. For $t > t_*$, with $\varkappa = \sqrt{Q^2 - m^2}$ and $W(t) = \int_{t_*}^t \varkappa dt' = H^{-1}[\sqrt{Q^2 - m^2} - m \arccos(m/Q)]$, the WKB growth is $d \ln(u^\dagger u)/dt = 2\varkappa$ at leading order and $2\varkappa - m^2/\varkappa^2$ at first adiabatic order.

10.3 Initial data and parameters

$q \in \{0.05, 0.1\}$, so $t_* = 2.99573$ and 2.30259 ; integration from $t = 0$ to $t_* + 3$ with 600 output intervals. The initial spinor is the positive-energy eigenvector of $h(0)$ ($E_0 = \sqrt{m^2 - q^2} = 0.998749$ and 0.994987) that is also an eigenvector of C with eigenvalue ± 1 (C commutes with $h(t)$ for momentum along x_5), so $u^\dagger Bu = \pm E_0/m \neq 0$. Four runs. The WKB comparison uses the window $[t_* + 1.5, t_* + 3]$, away from the turning point.

10.4 Solver settings and statistics

Adams + fixed point, rtol 10^{-10} , atol 10^{-12} , max_step 0.02: 2548 steps and 4800 right-hand-side evaluations.

10.5 Results

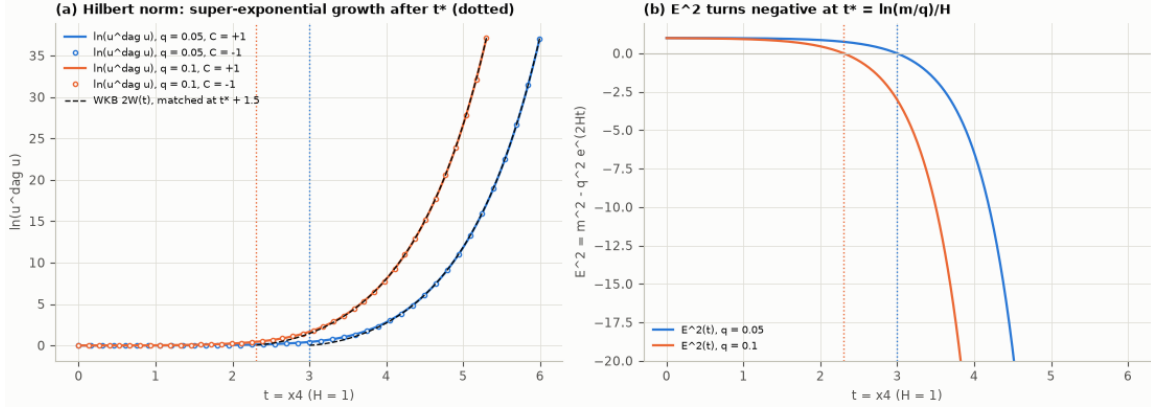


Figure 22: EXP-5 (a) $\ln(u^\dagger u)$ for both momenta and both C signs with the WKB curve $2W(t)$, and (b) $E^2(t)$, which turns negative at $t_* = \ln(m/q)/H$ (dotted).

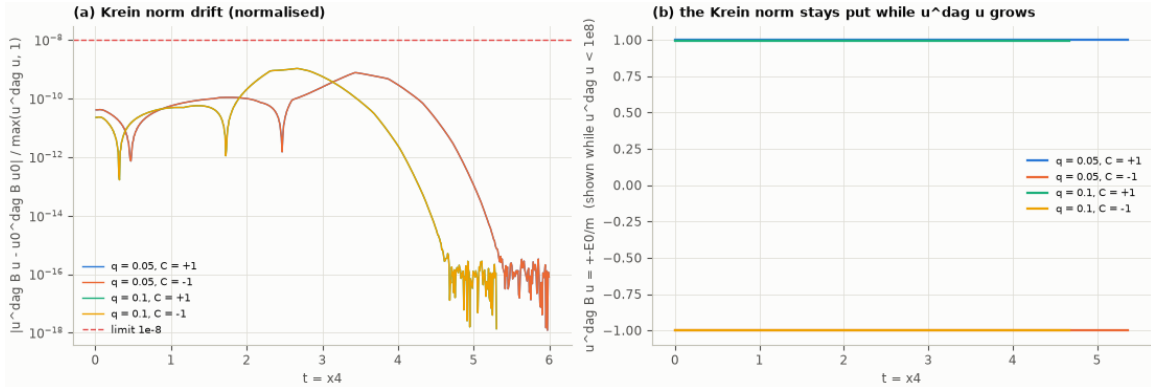


Figure 23: EXP-5 (a) the Krein-norm drift normalised by $\max(u^\dagger u, 1)$, below 10^{-8} throughout, and (b) the Krein norm, which stays at its initial value while $u^\dagger u$ grows.

- $u^\dagger u$ reaches 1.258×10^{16} ($q = 0.05$) and 1.313×10^{16} ($q = 0.1$). The growth rate $d\ln(u^\dagger u)/dt$ rises from 2.37 ($q = 0.05$) and 2.40 ($q = 0.1$) early to 25.3 late, ratios 10.7 and 10.6: super-exponential growth.
- Over the window, $\Delta \ln(u^\dagger u) = 30.9922$ against the leading WKB value 31.0241 (relative deviation 1.03×10^{-3}) and the first-order value 30.99986 (2.47×10^{-4}) for $q = 0.05$; 30.9455 against 30.9770 and 30.9530 for $q = 0.1$. The late rate matches $2\kappa - m^2/\kappa^2$ to 4.6×10^{-6} .
- The Krein norm is conserved: normalised drift at most 1.08×10^{-9} , absolute drift up to t_* at most 1.28×10^{-9} . At the end the absolute drift is 1.005 because $u^\dagger B u$ is then an $O(1)$ difference of terms of order 10^{16} , the float64 cancellation floor.

10.6 Verification

The Rust program passes 7 of 7 self-checks and the EXP-5 checker 21 of 21 (finite-difference residual ratio 0.71; independent RK4 reference 3.34×10^{-8} relative, refined 2.28×10^{-9} ; closed-form WKB against quadrature 1.1×10^{-14} ; repeat-run byte identity). The Mathematica notebook verifies $h^2 = (m^2 - Q^2)I$, $h^\dagger B = Bh$ and that h is not Hermitian; NDSolve agrees to 3.34×10^{-8} relative and a WorkingPrecision-32 run agrees with machine precision to 1.8×10^{-14} .

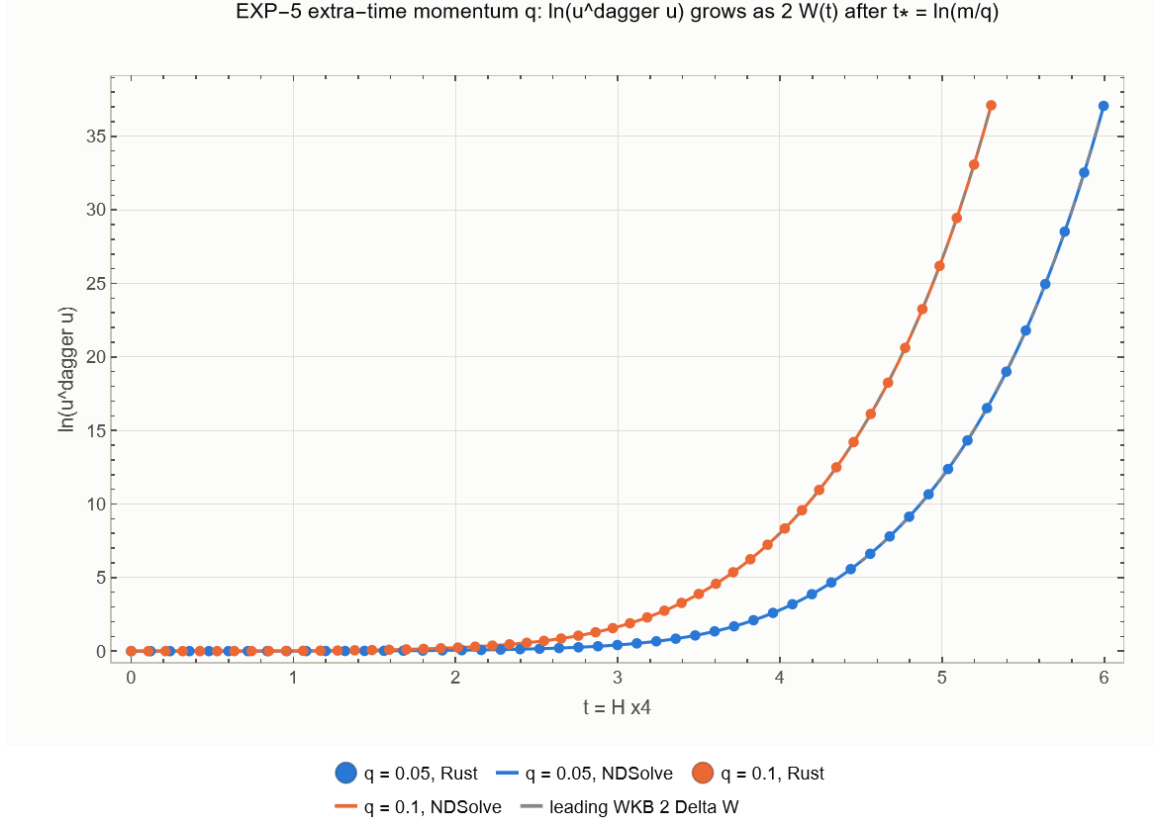


Figure 24: Mathematica cross-check of the Hilbert-norm growth: Rust (dots), NDSolve (lines) and the leading WKB curve.

10.7 Physical interpretation

A mode with extra-time momentum stops oscillating and grows once the extra-time scale factor has deflated enough. The growth conserves the indefinite Krein norm, so it is a growing positive-norm part and a growing negative-norm part together: the ultrahyperbolic ill-posedness of the (4,4) signature. No positive Fock space exists for these modes. Any cosmology built on this field must exclude them, and every result of EXP-1 to EXP-4 is restricted to the good sector. This restriction is imposed, not derived; whether a dynamical mechanism suppresses the extra-time sector is open.

11 Independent cross-checks: the Jupyter and Mathematica notebooks

11.1 The Jupyter notebook

The Jupyter notebook and its tools are these files:

<code>notebooks/dirac16complex_dark_sector.ipynb</code>	the executed notebook
<code>notebooks/build_dirac16complex_notebook.py</code>	writes the notebook
<code>notebooks/run_notebook.py</code>	headless executor
<code>notebooks/check_notebook.py</code>	auditor

The notebook has 23 cells (15 markdown, 8 code; kernel python3). It runs every subcommand of the Rust program into the git-ignored build/notebook-run and requires all 62 files written by the program to be byte-identical to the committed ones and the 3 files of the EXP-3 analysis to agree with them value by value (relative 10^{-6} , absolute 10^{-8}), because numpy versions and builds differ in the last digits of those files and, in the flat directions of the Nelder-Mead distance fits, in the fitted parameters by up to about 10^{-7} relative (measured between Windows and Ubuntu; the fit residuals agree to about 10^{-13}). It prints no folder name, so its output does not depend on what the clone is called. It then recomputes the physics from the raw spinor columns with the fixture's gamma matrices, for example the exact EXP-1 propagator (3.12×10^{-9}) and its own Gauss-Legendre quadrature of the EXP-3 comoving distance (1.7×10^{-9} from CVODE), and writes the 17 figures with fixed metadata. A gauntlet of 71 assertions covers every Rust, checker and analysis check, the Stage-1 results (153 of 153 checks), the fixture hash across all reports, the notebook's own recomputations and the 72 numbers quoted in its markdown. The notebook was executed headless by `run_notebook.py` and by `python -m nbconvert`. The auditor checked both executed copies (verdict SUCCESS, identical gauntlet results and figure hashes) and wrote notebook-report.json.

11.2 The Mathematica notebook

The Mathematica notebook and its two scripts are these files:

<code>notebooks/Dirac16ComplexDarkSector.nb</code>	the notebook
<code>scripts/build_dirac16complex_mathematica_notebook.wls</code>	writes it deterministically
<code>scripts/verify_dirac16complex_mathematica_notebook.wls</code>	evaluates it headless

The verifier fails on any message, any failed evaluation or any false check. The notebook loads the Stage-1 packages `Dirac16ComplexAlgebra.wl` and `Dirac16ComplexGeometry.wl` from `wolfram/`. It reads the gamma matrices, C , γ^8 and B back from `src/generated.rs` of the study and requires them to equal the package matrices exactly. It derives every reduced system from $\gamma^\mu D_\mu \Psi = M_{\text{eff}} \Psi$ with the package geometry and compares it exactly with the Rust form. It runs the binary with `RunProcess` (print-config: exit code 0, last line SUCCESS, fixture hash and tolerances equal to the summaries). All 49 checks are true. The independent integrations (NDSolve against Rust unless stated):

Quantity compared	Maximum deviation
EXP-1 spinor state	2.92×10^{-9}
EXP-1 exact-propagator error, Rust	6.25×10^{-9}
EXP-1 exact-propagator error, NDSolve	4.3×10^{-14}
EXP-2 $\ln h_i$	3.16×10^{-8}
EXP-2 H_i/Θ	3.24×10^{-8}
EXP-2 $\ln h_i$ against the closed form, NDSolve	2.7×10^{-11}
EXP-2 spinor, raw	1.08×10^{-7}
EXP-2 spinor, phase aligned	4.2×10^{-10}
EXP-2 spinor against the exact solution, Rust	2.24×10^{-7}
EXP-2 the same, NDSolve	7.7×10^{-10}
EXP-3 spinor	2.12×10^{-8}
EXP-3 $\ln \sigma$	1.95×10^{-14}
EXP-3 spinor against the exact solution, Rust	5.48×10^{-8}
EXP-3 the same, NDSolve	2.0×10^{-12}
EXP-4 thermal modes, raw	2.62×10^{-6}
EXP-4 thermal modes, phase aligned	5.9×10^{-8}
EXP-4 NDSolve Adams against Runge-Kutta	6.6×10^{-9}
EXP-4 kinetic theory, ρ (relative)	1.98×10^{-8}
EXP-4 kinetic theory, p (relative)	1.87×10^{-5}
EXP-4 pair modes, phase aligned	7.6×10^{-9}
EXP-4 pair $ \beta ^2$ (relative)	1.9×10^{-9}
EXP-5 state (relative)	3.34×10^{-8}
EXP-5 machine against 32-digit precision	1.8×10^{-14}

The finite-difference right-hand-side test uses 9-point stencils; rows the output grid cannot resolve (rate times stencil width above 4) are excluded and counted, and EXP-4, whose sampling is too sparse, uses one-interval NDSolve propagation instead (1.3×10^{-10}). Two environment details are documented in the notebook: the first PNG export in a headless kernel raises one named message about an optional packet, which is silenced by name while the file is written, and the high-precision runs use exact coefficients.

12 Synthesis: pressure, energy density, dark matter and dark energy

12.1 How pressure and energy density behave and change

- **Free condensate, any background (single-mode mean field):** $\rho = mS \propto 1/V$ and $p = 0$: dust, whatever the geometry does, up to the Fermi-degeneracy pressure of a Pauli-consistent state, which was not computed (Section 4.6).
- **Repulsive condensate ($\lambda > 0$):** $\rho = mS + \frac{\lambda}{2}S^2$ and $p = \frac{\lambda}{2}S^2 > 0$; $w \rightarrow 1$ (stiff) at small V and $w \rightarrow 0$ at large V (EXP-1 in the pointwise mean field, EXP-2).
- **Attractive condensate ($\lambda < 0$):** $p < 0$ and $w \rightarrow 0$ at large V ; where $M_{\text{eff}} < 0$ the equations

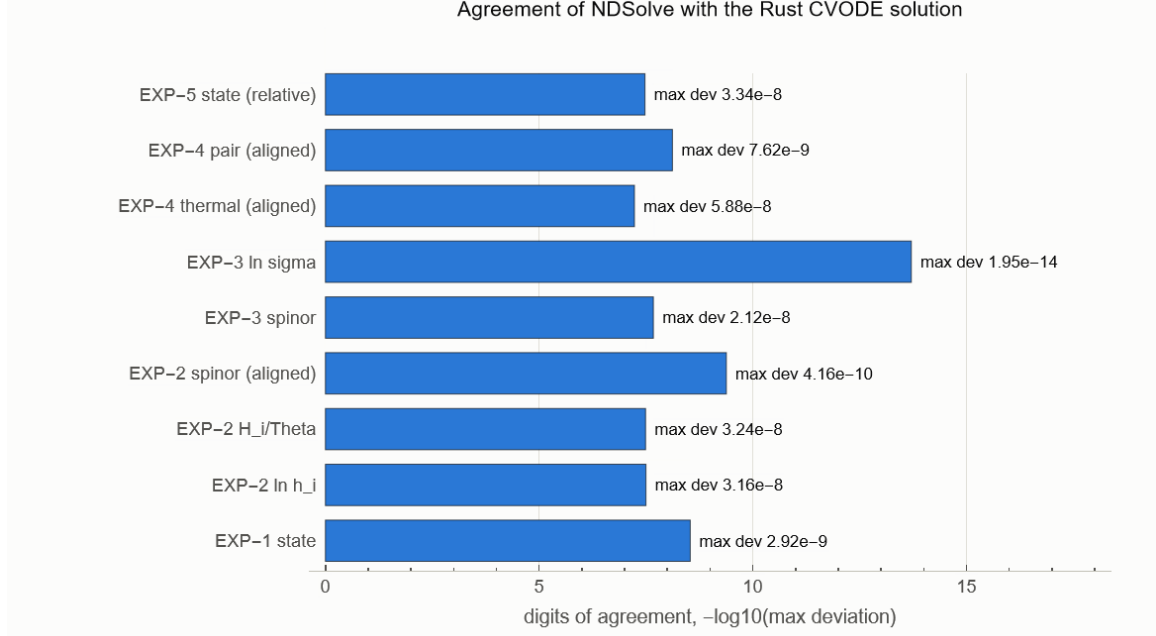


Figure 25: Agreement of NDSolve with the Rust CVODE solutions, as digits of agreement $-\log_{10}$ of the maximum deviation, for each experiment.

give phantom w and, at still smaller V , $\rho < 0$, but there the occupied mode is a negative-energy level and these are mean-field artefacts (EXP-2, EXP-3).

- **Primordial field:** V is constant, so ρ and p are frozen; dust or slightly stiff for the positive-energy eigenstates (EXP-1). The $2E$ oscillation of the mixed control runs is a first-quantised interference effect, not a property of the field.
- **Thermal gas:** $\rho \propto a^{-4}$ early and $\propto a^{-3}$ late; w falls from 0.3329 to 0.0359 (truncated grid; 0.0361 without the momentum cut) as the momenta redshift (EXP-4).
- **Created pairs:** the final spectrum has w between 0.24 and 0.31 when evaluated at $a = 1$ and about 10^{-4} at the end of the run, and $\rho a^3 \rightarrow m n a^3$ (EXP-4); most quanta are produced in the radiation era, when H is of order m .
- **Extra-time modes:** no positive norm and super-exponential growth after t_* ; ρ and p are not defined in a positive Fock space (EXP-5).

Three mechanisms control everything. **Dilution:** SV is constant, so the mass term behaves like dust in the 7-volume and the four-fermion term like a stiff fluid in the 7-volume. **Redshift:** the pressure of quanta is momentum flux and falls faster than their energy. **Geometry:** what a 3-space observer infers depends on how the 7-volume relates to the 3-space volume, $w_{\text{eff}} = -1 + \Theta(1 + w)/(3H_a)$. In EXP-1 $\Theta = 0$, in EXP-2 $\Theta \rightarrow 7H_a$, and in EXP-3 and EXP-4 $\Theta = 3H_a$ by assumption.

12.2 Dark matter

What behaves like dark matter.

- A Pauli-consistent gas of quanta ($f \leq 1$, $8 + 8$ states per momentum, $k_0 = 0$) has a time-varying equation of state, relativistic to dust: $w = 0.3329$ at $T = 10m$ and 0.0359 after a hundredfold expansion on the truncated grid (0.0361 without the cut), in agreement with kinetic theory on the same grid; $\text{KE}_H/\rho = 3w$ falls from 1 to about 0.11 while the rest-mass fraction PE_H/ρ rises (EXP-4a). This is the dark-matter-like result that does not depend on the single-mode picture.
- Gravitational pair creation by the de Sitter to radiation transition gives $na^3 = 1.454 \times 10^{-3}$, 4.990×10^{-3} , 4.412×10^{-3} and $2.421 \times 10^{-3} H_{\text{inf}}^3$ for $m/H_{\text{inf}} = 0.1, 0.5, 1$ and 2 (final spectrum in the first-order adiabatic basis, $k \leq 40$), and nothing for $m = 0$ (EXP-4b); most of it happens in the radiation era, when H is of order m . The produced gas is dust ($w \approx 10^{-4}$) by the end of each run.
- The free mean-field condensate is dust in every background: $w = 0$, $\text{KE}_L = \text{PE}_L = \rho/2$, $\text{KE}_H = 0$, $\text{PE}_H = \rho$ (EXP-1 with $K = 0$, EXP-2 and EXP-3 with $x_0 = 0$). That is a property of the single-mode picture: a Pauli-consistent state at the same density is dust only up to a Fermi-degeneracy pressure, negligible when $k_F(n) \ll m$ (Section 4.6), which is plausible for EXP-3 with a physical mass of order eV or more but not guaranteed for the EXP-2 parameters.
- Any repulsive or attractive condensate becomes dust at large volume (EXP-2, EXP-3), because the interaction energy dilutes faster.

What is not established.

1. **Abundance.** No relic density $\Omega_{\text{DM}} h^2$ was computed. The yields are per comoving volume in units of H_{inf}^3 at the end of de Sitter; converting them requires physical values of m and H_{inf} and the reheating history.
2. **Darkness.** The framework has no coupling to the standard model, so the quanta are dark by construction; no bounds on interactions, annihilation through the λ term or decays were derived.
3. **Stabilised extra dimensions.** In self-consistent 8D gravity (EXP-2) all seven directions end up expanding equally, and a 3-space observer sees the dust dilute like a^{-7} ($w_{\text{eff}} \rightarrow 4/3$). Dust-like dilution in 3-space, which dark matter needs, requires static extra dimensions; that was assumed in EXP-3 and EXP-4, not derived.
4. **The extra-time sector** must be excluded by hand (EXP-5).
5. **Structure formation**, free streaming of the warm phase and clustering were not computed.
6. **The hidden-space momentum** k_0 was set to zero in EXP-4 (a compact hidden dimension with a large Kaluza-Klein gap, or by hand); with it the relativistic pressure would be $\rho/4$.

Answer on dark matter. There is a qualitative connection: in the good sector, with stabilised extra dimensions, `dirac16complex` quanta are pressureless matter at late times (and so are free condensates in the single-mode mean field), and expansion produces the quanta gravitationally. The connection is an equation of state and a production mechanism, not a dark-matter model, and nothing computed here predicts the observed dark-matter density.

12.3 Dark energy: which mechanisms give negative pressure or $w < -1/3$

1. **Attractive four-fermion interaction** ($\lambda < 0$). The only mechanism with genuinely negative pressure, $p = \frac{\lambda}{2}S^2 < 0$. $w = x_0\sigma/(1 + x_0\sigma)$ lies in $(-1, 0)$ at low density. At higher density M_{eff} passes through zero, and beyond that point the equations give $w < -1$ with $\rho > 0$ ($\text{KE}_L = \frac{1}{2}SM_{\text{eff}} < 0$), then $\rho < 0$, and in EXP-3 a bounce (EXP-2: phantom for $0.4 < V/V_0 < 0.8$ and $\rho < 0$ below 0.4). **This phantom crossing is a mean-field artefact, not an established property of the field:** in all those rows the single occupied mode is a negative-energy eigenvector of the instantaneous h ($s(u) = 1$ while $M_{\text{eff}} < 0$), outside the domain of the expectation-value rule, and no state of positive-energy quanta has $\text{KE}_L < 0$ (Section 4.6).
2. **Frozen density in the primordial field.** In EXP-1 the 7-volume is constant ($\Theta = 0$), so the density does not dilute while 3-space inflates. For a 3-space observer that is the dilution of a cosmological constant: $w_{\text{eff}} = -1 + \Theta(1 + w)/(3H_a) = -1$ (derived here from the EXP-2 definition and the EXP-1 background, not a separate machine check). The actual pressure is zero or positive, and the geometry is not self-consistent: 8D Einstein gravity would need $\rho_{\text{req}} \leq -21$ (Section 6.5).
3. **Deflating extra times** (EXP-3 variant). With $c \propto a^{-\gamma}$, $\sigma \propto a^{-n}$, $n = 3(1 - \gamma) < 3$, the density dilutes more slowly than a^{-3} . For $\gamma = 0.875181$ ($n = 0.374457$, $x_0 = -0.4626545$) the tangent CPL parameters of p/ρ at $a = 1$ equal the Unite fit values. The Unite values are fit parameters over the data range, not a tangent; the like-for-like comparison, the least-squares CPL fit of the variant's own p/ρ on $a \in [1/3.26, 1]$, gives $(w_0, w_a) = (-0.640, -1.98)$ (rms residual 0.154), far from Unite. Distances measure w_{eff} with tangent $(-0.9827, -0.0749)$. Section 12.5 lists why it fails.

The Lagrangian has no cosmological-constant term ($U(0) = 0$ is forced for a Grassmann field), and a condensate has $w = -1$ only at the instant $M_{\text{eff}} = 0$.

12.4 Quantitative comparison with Unite

The fitted and tangent parameters per scenario. "Requested" fits are the programme's: least-squares CPL fit of $w(a)$ on $a \in [1/3.26, 1]$, and CPL and constant- w fits of $\text{DM}(z)$ on $z \in [0.01, 2.26]$ with the offset profiled. For $x_0 < -0.028863$ the $w(a)$ fit does not exist (w has a pole at $a = |x_0|^{1/3}$ inside the range) and for $x_0 < -0.041024$ the $\text{DM}(z)$ fits do not exist (the bounce lies at $z_b < 2.26$). The restricted fits use the largest range on which each model is defined ($w \geq -3$; $z < z_b$); they are supplementary and not the programme's fits.

Tangent parameters and the status of the requested fits:

x_0	tangent w_0	tangent w_a	requested fits
-0.462654	-0.861	-4.807	undefined
-0.433107	-0.764	-4.043	undefined
-0.3	-0.4286	-1.837	undefined
-0.2	-0.25	-0.9375	undefined
0	0	0	(0, 0) and $w = 0$

Restricted least-squares CPL fit of $w(a)$ (supplementary):

x_0	range of a	w_0	w_a
-0.462654	[0.851, 1]	-0.605	-12.83
-0.433107	[0.833, 1]	-0.485	-11.68
-0.3	[0.737, 1]	-0.074	-7.583
-0.2	[0.644, 1]	0.130	-5.267
0	[0.307, 1]	0	0

Restricted CPL fit of $DM(z)$ with the offset profiled (supplementary):

x_0	range of z	(w_0, w_a)	rms residual
-0.462654	[0.01, 0.388]	(2.106, -48.18)	0.078 mag
-0.433107	[0.01, 0.423]	(1.846, -39.84)	0.074 mag
-0.3	[0.01, 0.633]	(1.164, -18.11)	0.062 mag
-0.2	[0.01, 0.890]	(0.871, -9.943)	0.056 mag
0	[0.01, 2.26]	(0, 0)	0

Restricted constant- w fit of $DM(z)$ and the largest distance-modulus difference from the Unite CPL model over the same range (same H_0):

x_0	constant w	rms residual	$\max \Delta DM $
-0.462654	-2.764	0.120 mag	0.832 mag
-0.433107	-2.478	0.115 mag	0.785 mag
-0.3	-1.524	0.099 mag	0.549 mag
-0.2	-1.012	0.090 mag	0.329 mag
0	0	0	0.795 mag

References and the deflation variant:

Reference or variant	(w_0, w_a)	constant w
Unite CPL (PDF)	(-0.861, -0.60)	none
Unite Λ CDM (PDF)	none	-0.764
best constant w for the Unite CPL distances, $\Omega_m = 0.305$ fixed	none	-1.0277
the same with the offset fixed at 0	none	-0.9747
the same on a log-uniform z grid	none	-0.9790
the same with Ω_m free (best $\Omega_m = 0.2777$)	none	-0.9115

Reference or variant	(w_0, w_a)	constant w
the same, Ω_m free, offset fixed at 0	none	-0.8788
the same, Ω_m free, log-uniform z grid	none	-0.8721
Λ CDM	$(-1, 0)$	-1
deflation variant, tangent of p/ρ	$(-0.861, -0.60)$	none
deflation variant, p/ρ fit on $a \in [1/3.26, 1]$	$(-0.640, -1.98)$	none
deflation variant, tangent of w_{eff}	$(-0.9827, -0.0749)$	none
deflation variant, $w_{\text{eff}}(a)$ fit	$(-0.955, -0.247)$	none
deflation variant, $\text{DM}(z)$ fits	$(-0.9707, -0.1611)$	-1.0159

For the Unite CPL, $w_0 + w_a = -1.461$ and $w = -1$ at $a = 0.768$. The Unite CPL distances used as the target always have $\Omega_m = 0.305$, the value assumed by the programme; the PDF gives no Ω_m . The tangent of the deflation variant's p/ρ equals the Unite values by construction; its like-for-like fit $(-0.640, -1.98)$ does not.

The scan over $x_0 \in [-0.49, 0]$ in steps of 0.001 (491 models, fits_scan.csv) finds $x_0 = -0.4626545$ for $w_0 = -0.861$, with tangent $w_a = -4.807$, and $x_0 = -0.4331066$ for $w_0 = -0.764$.

Reading the tables.

- Matching today's Unite value $w_0 = -0.861$ forces $w_a = -4.81$: the condensate evolves about eight times faster than the Unite fit. Its effective mass vanishes and $w = -1$ at $z = 0.026$ (the Unite CPL crosses -1 at $a = 0.768$), and beyond that redshift the mean field is outside its valid regime; the equations there give negative energy density beyond $z = 0.293$ and a bounce at $z = 0.388$. Either way the model provides no description of the universe beyond $z = 0.026$, so it cannot be confronted with supernovae up to $z = 2.26$, let alone with the CMB.
- The fits that the comparison was meant to use do not exist for any $x_0 < -0.041$. The restricted fits give values such as $(w_0, w_a) = (2.1, -48)$ with residuals of 0.06 to 0.08 mag, which only show that the CPL form does not describe these models.
- **The benchmark $w = -0.764$.** The model reproduces it only as today's value ($x_0 = -0.433107$), with $w_a = -4.04$; a constant- w fit to that model's own distances gives -2.48 . How -0.764 relates to the Unite CPL curve depends on Ω_m , an assumed input here. With $\Omega_m = 0.305$ fixed and noise-free equal weights, -0.764 is not a constant- w projection of the Unite CPL curve: the best constant w for the CPL distances is -1.028 (uniform z), -0.975 (offset zero) or -0.979 (log-uniform z). An SN-only wCDM fit, which is what the PDF says -0.764 is, leaves Ω_m free along the strong w - Ω_m degeneracy. With Ω_m free the noise-free projection is -0.9115 ($\Omega_m = 0.2777$, rms residual 0.0021 mag instead of 0.0077 mag), -0.8788 (offset zero) or -0.8721 (log-uniform z): about -0.9 , not -0.764 , but most of the gap to -0.764 with Ω_m fixed was an Ω_m effect. The remaining difference cannot be judged without the Unite likelihood, its errors and its Ω_m .

- Only the deflation variant comes close to Unite in distances ($\max |\Delta \text{DM}| = 0.021$ mag, 0.011 mag with the offset profiled), and its distances look like a mildly evolving $w \approx -0.97$ to -1.02 , not like the Unite CPL; its own p/ρ fits to $(-0.640, -1.98)$, and it fails for other reasons (Section 12.5).

12.5 Where the model fails

1. **Rapid evolution.** $w_a = 3x_0/(1+x_0)^2$ is fixed by w_0 ; at the Unite w_0 it is -4.81 instead of -0.60 .
2. **The mean field leaves its domain.** Going back in time M_{eff} vanishes at $z = 0.026$ for the Unite-tuned model; beyond it the single occupied mode is a negative-energy level, and the phantom epoch, the negative energy density beyond the zero of ρ_ψ ($z = 0.293$) and the bounce ($z = 0.388$) that the equations give are artefacts of that approximation. Either way the model gives no matter era and no early universe.
3. **Sound speed.** $c_s^2 = dp/d\rho = 2x_0\sigma/(1+2x_0\sigma)$ is negative today for every attractive model ($-12.39, -6.47, -1.50, -0.667$). In a fluid description that is a gradient instability of perturbations (not computed); at the phantom crossing $1+2x_0\sigma = 0$ the closed form diverges.
4. **The 8D Einstein bound.** With $\rho \geq 0$, $w \geq -1$ and $H_a, \Theta > 0$, a 3-space observer cannot infer $w_{\text{eff}} < -1 + (1+w)/\sqrt{3}$ (EXP-2); for dust -0.42265 . The deflation variant (a condensate with $\rho > 0$ and $w \geq -1$ today) has $w_{\text{eff}} \approx -0.98$ and therefore needs negative 8D energy density: $\kappa\rho_8 = 3H_a^2(1-3\gamma+\gamma^2) = -2.579H_a^2$ for $\gamma = 0.875$; sustained deflation with γ between $(3-\sqrt{5})/2 = 0.382$ and $(3+\sqrt{5})/2 = 2.618$ always does. EXP-2 shows, moreover, that the extra times turn to expansion by themselves.
5. **Varying Newton constant.** $G_N = G_8/V_{\text{extra}} \propto a^{3\gamma}$ gives $\dot{G}/G = 3\gamma H_0 = 2.63H_0$, about 1.8×10^{-10} per year for $H_0 = 67.4$ km/s/Mpc, some 1800 times the order-of-magnitude lunar-laser-ranging bound of 10^{-13} per year, and $G_N(z=1)/G_N(0) = 0.162$. The constant- G_N Friedmann equation used in EXP-3 is itself inconsistent with this.
6. **4D energy conservation.** For $n \neq 3$ the condensate exchanges energy with the extra dimensions: $d\rho/dN + 3(\rho+p) = 0.254$ today in units of $3H_0^2/\kappa_4$. Distances then measure w_{eff} , not p/ρ .
7. **The deflation variant's own history.** Its ρ_ψ is negative beyond $z = 6.83$, and its p/ρ crosses $w = -1$ at $z = 0.230$; the CPL fit of that p/ρ over the data range is $(-0.640, -1.98)$, not the Unite values.
8. **Krein structure and the extra-time instability.** The canonical state space is indefinite; only the good sector has a positive Fock space, and modes with extra-time momentum grow super-exponentially (EXP-5). The restriction is imposed by hand.
9. **Homogeneous mean field only.** One c-number mode stands for the condensate, at Hartree level and without the Pauli constraint on its occupation; quantum corrections, exchange terms, the stability of the condensate and inhomogeneities were not computed.
10. **No perturbations and no data likelihood.** Nothing here is a fit to supernova, BAO or CMB data; the distance comparisons use noise-free curves with equal weights, and Ω_m is fixed at the assumed 0.305 except in the Ω_m -free projections of Section 12.4.

12.6 The answer

- **Dark matter: a qualified yes.** In the good sector, with stabilised extra dimensions and no hidden-space momentum, `dirac16complex` behaves like cold matter: a Pauli-consistent thermal gas turns from radiation into dust, the transition from inflation to radiation creates quanta for every $m > 0$ (mostly when H is of order m), and free condensates are dust in the single-mode mean field (up to a Fermi-degeneracy pressure that was not computed). This is a dark-matter-like equation of state and a gravitational production mechanism. The abundance, the absence of standard-model couplings, the stabilisation of the extra dimensions and the perturbations are not established, so it is not a dark-matter model.
- **Dark energy: no, within everything computed.** The only negative pressure comes from an attractive four-fermion interaction. Tuned to Unite’s $w_0 = -0.861$ it evolves eight times too fast ($w_a = -4.81$ against -0.60), has a negative sound speed squared, and its effective mass vanishes at $z = 0.026$, where the single mean-field mode stops being a positive-energy level. The phantom crossing, the negative energy density at earlier times and the bounce that the equations then give are mean-field artefacts, not established properties: no state of positive-energy quanta of this Lagrangian is phantom. The frozen density of the primordial field mimics a cosmological constant only in its dilution, in a geometry that needs negative energy. Deflating extra times give p/ρ the Unite values as its tangent at $a = 1$, but its CPL fit over the data range is $(-0.640, -1.98)$; they violate energy conservation in 4D and the 8D energy bound, and they make G_N vary about 1800 times faster than the lunar-laser-ranging bound allows. The framework reproduces neither Unite’s (w_0, w_a) nor $w = -0.764$ consistently.

13 Verification summary

Experiment	Rust self-checks	Python checker	Mathematica
EXP-1	10 of 10	25 of 25	6 of 6
EXP-2	15 of 15	34 of 34	5 of 5
EXP-3	14 of 14	31 of 31	4 of 4
EXP-4	23 of 23	51 of 51	5 of 5
EXP-5	7 of 7	21 of 21	4 of 4
total	69 of 69	162 of 162	24 of 24

Every checker includes repeat-run byte identity and refined-tolerance convergence; for the EXP-2 spinor phase the refined run confirms the identified rounding floor instead of a smaller error. In addition: the EXP-3 analysis, 10 of 10 checks; the Jupyter notebook’s gauntlet, 71 of 71 assertions in both executions, with 62 of 62 freshly written program files byte-identical and 3 of 3 analysis files numerically equal to the committed ones; and the remaining 25 of the Mathematica notebook’s 49 checks (10 algebra, 7 Rust constants, 6 for the binary, 1 for the quoted parameters and 1 for the figures). Every report records the fixture sha256 of Section 14. Where the numerical programme’s a priori expectations were contradicted by measurement, the checks test the measured truth and the reports record the deviation: in EXP-1 the pressures are frozen only for eigenstates; in EXP-2 the bound $\Theta > \sqrt{3}H_a$ holds only where $\rho \geq 0$, and the phantom and negative-energy rows lie outside the domain of the mean field; in EXP-3 the backward range ends at the bounce and the requested

fits do not exist; in EXP-4 the per-mode $|\beta|^2 \leq 10^{-6}$ is unattainable at $H_i/m = 0.05$ for a sudden start, and the pressure carries the free-wave interference; in EXP-5 the absolute Krein drift at late times is the float64 floor.

14 Files and hashes

sha256 of the inputs and outputs this document quotes. The fixture hash is the one recorded in every summary, every checker report and src/generated.rs; the hash of generated.rs itself was computed for this edition; the summaries, checker reports and fits.json carry the hashes recorded in numerics-summary.json. The publication test of this document recomputes all of them.

```
artifacts/dirac16complex/arbitrary-field/algebra-fixture.json
8b4f15462ca4d61ef6ec0e72f04c8a77d23ce8bcabc9b6ce19f921c90e02653b
studies/dirac16complex_cosmology/src/generated.rs
29631bcd9e737b5fe57a41ae2750c7637e31580974577d34b26ada45c560a3ba
artifacts/dirac16complex/numerics/exp1/summary.json
f8886625011a35c5ea31c89ffb623e4c6b0678adc300fc4b725179808601272a
artifacts/dirac16complex/numerics/exp1/python-check-report.json
840fc6fb8797fa66cd043a199357b599c6b2202cbeef426f0ed9058d318fafe4
artifacts/dirac16complex/numerics/exp2/summary.json
c8f7ecd95f7414f81e2569be7a41f2127d0c1cbd3a7a07b2c0dbcb667c43d5d0
artifacts/dirac16complex/numerics/exp2/python-check-report.json
b2d0c6d55cc0a3f6ebe0fcd3c6de0221964e32a9607c07e374398dcac4e70d5d
artifacts/dirac16complex/numerics/exp3/summary.json
7bfedf4bd5e5765c171dcc71896e4a9f481bfedfeafa6ddae04b924ca31081ee
artifacts/dirac16complex/numerics/exp3/python-check-report.json
86e92fbf565cc6498aed329fc6d5506e8749c6a31a24ec41f2f8f84048263406
artifacts/dirac16complex/numerics/exp3/fits.json
5cb8fe90bbc8c060d2b69d42b801657ab6a0bd56f81df20df95d567d9570412e
artifacts/dirac16complex/numerics/exp4/summary.json
361fb97728a36292b29c0434c431c9377501c8957299e34817fb4c1b47b53af1
artifacts/dirac16complex/numerics/exp4/python-check-report.json
10de81074dfd24eae9ecfa622d9be3b6122b72ba7ff6016ef30a9ed6c45d4b57
artifacts/dirac16complex/numerics/exp5/summary.json
bbddd4d4e0303c72d8d13494e2f4a54a730f0687fdb54a240156c7b0cfb1746
artifacts/dirac16complex/numerics/exp5/python-check-report.json
bf7d379b6cf41da6984c1a3490aec5ffedf968295ee52c2c06dcb4f5d8128e03
```

The sha256 of each of the 17 figures is recorded in notebook-report.json, and the figures of the Mathematica notebook are listed in mathematica-report.json. build_provenance_pdf.py records the sha256 of every figure of this document in its build report.

15 Reproduction

Run every command from the repository root. The engine (once per clone):

```
.\scripts\setup_solver.ps1 -Platform win11
bash scripts/setup_solver.sh
```

(the first in PowerShell, the second in Git Bash, macOS or Linux; the platform is detected or given as win11, macos or linux). The Python packages, at the versions with which the committed files were produced:

```
python -m pip install -r requirements-stage3.txt
```

With other numpy or matplotlib versions every check still passes, but the last digits of the analysis files and the checker reports can differ from the committed ones. Build and run all five experiments, then the EXP-3 analysis:

```
cargo build --release --manifest-path studies/dirac16complex_cosmology/Cargo.toml
studies/dirac16complex_cosmology/target/release/dirac16complex_cosmology all
python scripts/analyze_dirac16complex_exp3.py
```

The last line of the program is SUCCESS. The independent checkers, each with repeat-run byte identity and refined convergence (the scratch directories under build/ are git-ignored):

```
python scripts/check_dirac16complex_exp1.py --repeat build/repeat --refined build/refined
python scripts/check_dirac16complex_exp2.py --repeat build/repeat --refined build/refined
python scripts/check_dirac16complex_exp3.py --repeat build/repeat --refined build/refined
python scripts/check_dirac16complex_exp4.py --repeat build/repeat --refined build/refined
python scripts/check_dirac16complex_exp5.py --repeat build/repeat --refined build/refined
```

Each prints its check lines and exits 0 only if every check is true; it rewrites the committed python-check-report.json of its experiment.

The notebooks and this document:

```
python notebooks/run_notebook.py --quiet notebooks/dirac16complex_dark_sector.ipynb
python notebooks/check_notebook.py notebooks/dirac16complex_dark_sector.ipynb
wolframscript -file scripts/verify_dirac16complex_mathematica_notebook.wls
python -m unittest tests.test_d16c_numerics_publication -v
python scripts/build_provenance_pdf.py provenance/DIRAC16COMPLEX_DARK_SECTOR_NUMERICS.md
```

Running the Mathematica verifier rewrites mathematica-report.json and the figures under figures/mathematica (byte-identically). The PDF step builds this document twice (two builder runs, three pdflatex passes each, with the repository root as working directory so that the figures are found), requires warning-free logs and byte-identical PDFs, compares the result with the registered edition dirac16complex-dark-sector-numerics in provenance/pdf-specifications.json and prints provenance_pdf=OK. After an edit of this document the edition is registered again with

```
python scripts/build_provenance_pdf.py --register \
    provenance/DIRAC16COMPLEX_DARK_SECTOR_NUMERICS.md
```

(in PowerShell the two lines are written as one line), and the two sha256 pins in the publication test of this document are updated. The student guide of this stage, whose PDF lies next to this one in provenance/, gives every step in detail for Windows, macOS and Linux.

16 Limitations

1. **Backgrounds.** All experiments use homogeneous diagonal metrics. EXP-1 prescribes the primordial field, EXP-3 and EXP-4 prescribe 4D or 3-space FRW expansions with static extra dimensions, and only EXP-2 solves 8D Einstein gravity self-consistently. Einstein-Lovelock terms are not included.
2. **Mean field.** The condensate is one c-number rest mode with the expectation-value rule, at Hartree level. The rule holds only while that mode is a positive-energy level; where $M_{\text{eff}} < 0$ (the phantom, negative-energy and bounce regimes of EXP-2 and EXP-3) the results are mean-field artefacts. A macroscopic occupation of one momentum state violates the Pauli principle; the Fermi-degeneracy pressure of a Pauli-consistent state and the Fock exchange term were not computed. Its quantum stability, fluctuations and renormalisation are not studied. The gas and the pairs are free modes; the λ interaction between quanta is not included in EXP-4, and the hidden-space momentum k_0 is set to zero there.
3. **Good sector only.** Modes with extra-time momentum are excluded by hand because they are unstable (EXP-5); the Krein space is not given a physical interpretation beyond the good sector.
4. **Units and abundances.** The runs use m , H , H_0 or H_{inf} as units. The EXP-3 masses m/H_0 of 3 to 94 serve only to show phase independence, and no physical mass, relic abundance or inflation scale is predicted.
5. **Observational comparison.** The Unite numbers are as quoted in the reference e-mail, which cites no primary publication; they were not checked against one. No error bars, covariances or data were used. The ± 0.118 band of the $w(a)$ figure is an inferred approximation. $\Omega_m = 0.305$ is an assumed input (the PDF gives none); the distance fits are equal-weight fits of noise-free curves with it fixed, except the Ω_m -free constant- w projections of Section 12.4.
6. **Perturbations.** The sound speed is the adiabatic $dp/d\rho$ of the homogeneous solution; no perturbation equations, structure growth or CMB spectra were solved.
7. **Numerics.** The EXP-2 spinor phase is limited by the rounding of CVODE's internal time; the EXP-4 thermal phase error (at most 6.5×10^{-6} rad at $a = 100$ against the Magnus reference) is not reduced by the refined run, consistent with the same effect, and only phase-invariant quantities enter the results. The EXP-5 absolute Krein drift at the end is limited by float64 cancellation. The EXP-4 thermal gas is truncated at $k_{\text{max}} = 12T_i$ (2.4×10^{-3} of ρ at $a = 1$; $w(100) = 0.0359$ on the grid against 0.0361 without the cut), and the pair-creation grid at $k = 40$ (the analytic kink tail is reported as a correction). These floors are identified and reported, not removed.
8. **Scope of the answer.** "No viable dark energy" refers to the mechanisms available in this Lagrangian and these backgrounds. A different potential, non-homogeneous states, other couplings or a mechanism that stabilises the extra dimensions were not examined.